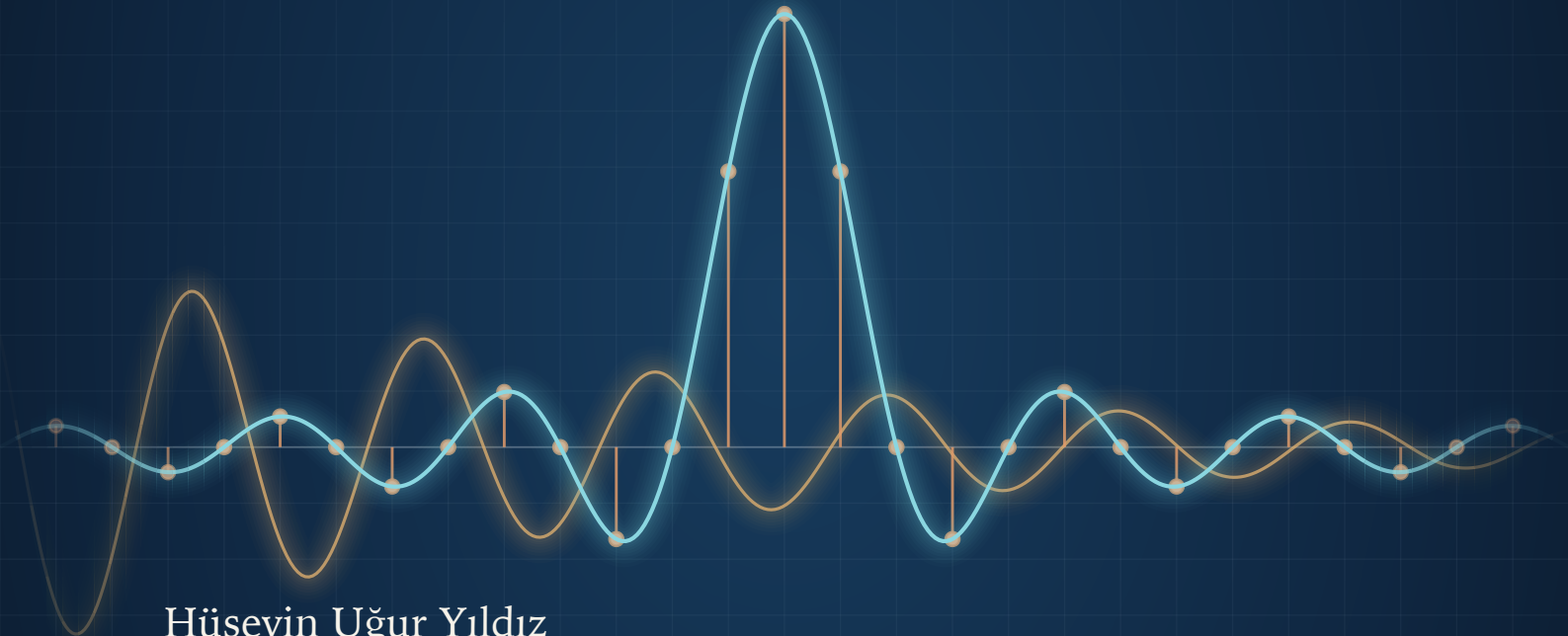




SIGNALS AND SYSTEMS

Signals, Systems and Frequency-Domain Analysis

Student Workbook



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Contents

Every question in the course, with no answer and no solution. Work each one on the page, then check it against the artifact or against the instructor edition.

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How to use it

The questions are in the order the course meets them, and each has a stable question number. Only the statement and its lettered parts are printed; the reasoning stays for you to supply. The question numbers are shared with every other edition, so D5-04 is the same question in the artifact, in this workbook and in the instructor solutions.

MODULE

1

Signal Foundations

30 questions on signal foundations. Write your reasoning in the space under each one.

D1-01

Decide whether each sequence below is periodic. Where it is, give the fundamental period N_0 .

$$(i) \ x[n] = \cos\left(\frac{5\pi}{8}n - \frac{\pi}{6}\right) \quad (ii) \ x[n] = e^{j3n} \quad (iii) \ x[n] = \sin\left(\frac{\pi}{6}n\right)$$

- a)** Apply the periodicity test to each sequence.
b) Give N_0 wherever it exists, and say why it does not exist otherwise.

D1-02

Let

$$x[n] = \cos\left(\frac{\pi}{5}n\right) + \sin\left(\frac{\pi}{3}n\right).$$

- a)** Show that each term is periodic and give its period.
b) Determine the fundamental period N_0 of $x[n]$ and the fundamental frequency.

D1-03

Decide whether each continuous-time signal is periodic. Where it is, give T_0 and ω_0 .

$$(i) \ x(t) = \cos(3t) + \sin(5t) \quad (ii) \ y(t) = \cos(t) + \cos(\sqrt{2}t)$$

- a)** Give the period of each term separately.
b) Decide whether the sum repeats, and give T_0 where it does.

D1-04

Let

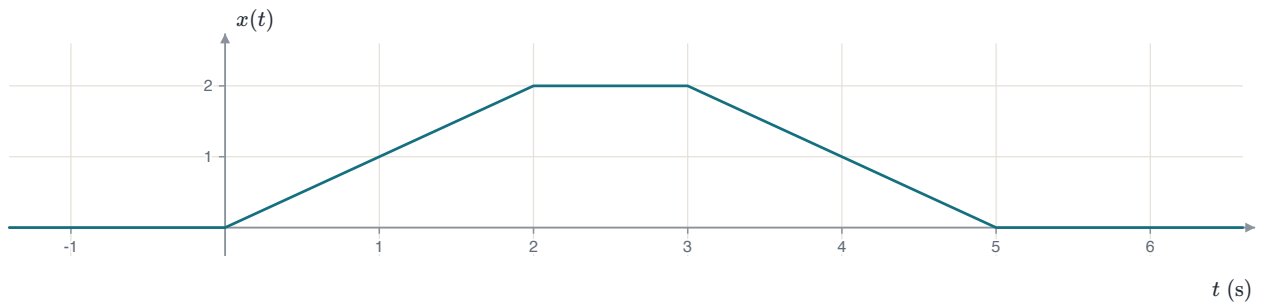
$$x(t) = 2e^{-0.3t} \cos(4t), \quad t \in \mathbb{R},$$

a sinusoid whose envelope is $\pm 2e^{-0.3t}$.

- State the period T_0 of the pure oscillation $y(t) = 2 \cos(4t)$, the case $r = 0$.
- Prove that $x(t)$ itself is not periodic, by testing the necessary condition $x(0) = x(T)$ for a candidate period $T > 0$.
- Explain why no signal whose envelope is strictly monotonic can be periodic.

D1-05

The signal $x(t)$ sketched below is zero outside $0 \leq t \leq 5$.



- Calculate the total energy E_∞ .
- Calculate the average power P_∞ and classify the signal.

D1-06

Consider the two discrete-time signals

$$x_1[n] = \left(\frac{2}{5}\right)^n u[n], \quad x_2[n] = 3u[n].$$

- Calculate E_∞ and P_∞ for $x_1[n]$.
- Calculate E_∞ and P_∞ for $x_2[n]$.
- Classify each signal.

D1-07

With the normalised convention $R = 1 \Omega$, consider

$$x_1(t) = e^{-5t}u(t) \text{ V}, \quad x_2(t) = 4 \cos(3t) \text{ V}.$$

- a) Calculate E_∞ for $x_1(t)$ and state P_∞ .
- b) Calculate P_∞ for $x_2(t)$ and state E_∞ .
- c) Starting from $P_T = \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$, show directly that a finite E_∞ forces $P_\infty = 0$.

D1-08

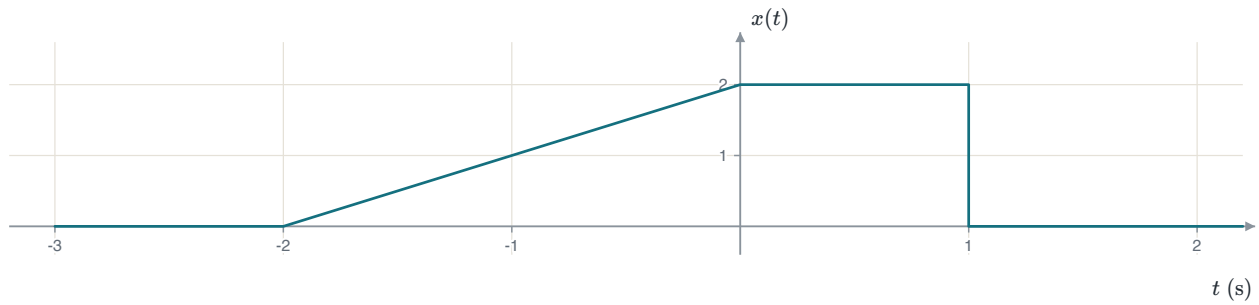
Let

$$x[n] = \begin{cases} 2^n, & n \leq -1, \\ 3, & n = 0, \\ \left(\frac{1}{2}\right)^n, & n \geq 1. \end{cases}$$

- a) List $x[n]$ for $-3 \leq n \leq 3$.
- b) Calculate E_∞ .
- c) State P_∞ and classify the signal.

D1-09

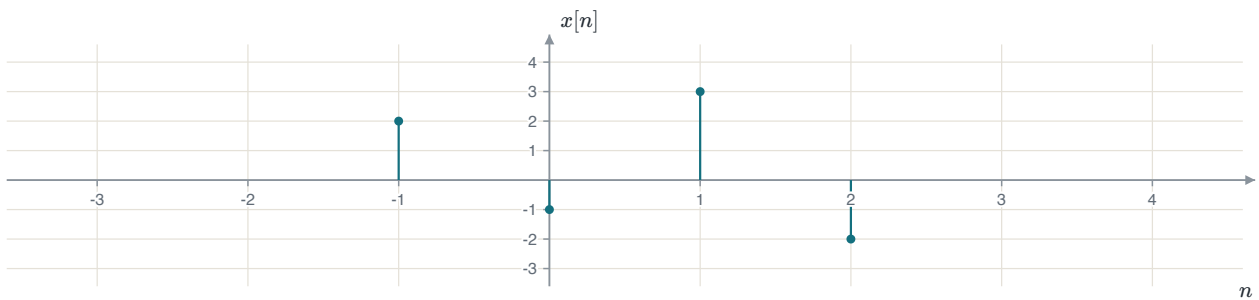
The signal $x(t)$ is sketched below: it rises linearly from 0 at $t = -2$ to 2 at $t = 0$, stays at 2 until $t = 1$, and is zero elsewhere.



- Plot $y(t) = x(2t + 2)$.
- State the support of $y(t)$ and check its width against the width of $x(t)$.

D1-10

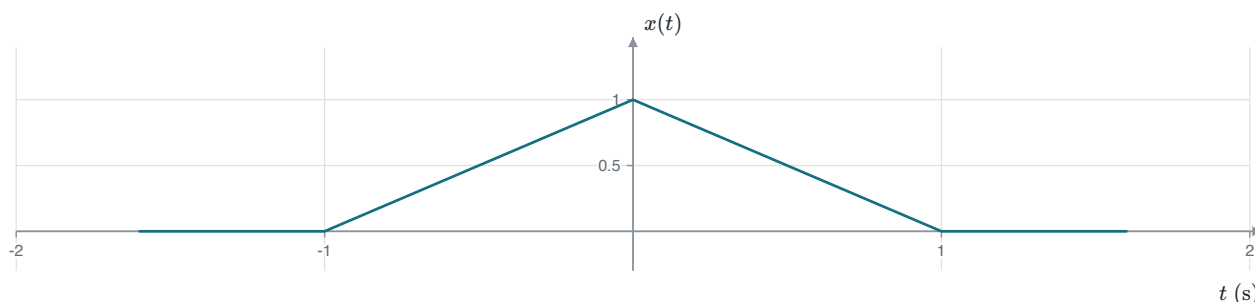
The sequence $x[n]$ is plotted below.



- Plot $y[n] = x[2 - n]$.
- Plot $z[n] = x[n + 1]$ and say which of the two operations changes the order of the samples.

D1-11

The signal $x(t)$ sketched below is a symmetric triangular pulse of height 1, zero outside $-1 \leq t \leq 1$.



- Plot $y(t) = x(1 - \frac{t}{3})$.
- State the support of $y(t)$ and check its width against the width of $x(t)$.
- Verify the location of the peak of $y(t)$ by direct substitution.

D1-12

A signal $x(t)$ is known only to satisfy $x(t) = 0$ outside $2 \leq t \leq 8$, with a single, unique maximum at $t = 5$. No other property of $x(t)$ is given. Let $y(t) = x(-3t + 6)$.

- Find the support of $y(t)$ as an interval, using inequalities on the argument of x .
- Find the value of t at which $y(t)$ reaches its maximum.
- A second transformation is applied, $z(t) = y(t - 4)$. Find the support of $z(t)$ from the support of $y(t)$. Then find it again from the support of $x(t)$ and compare the results.

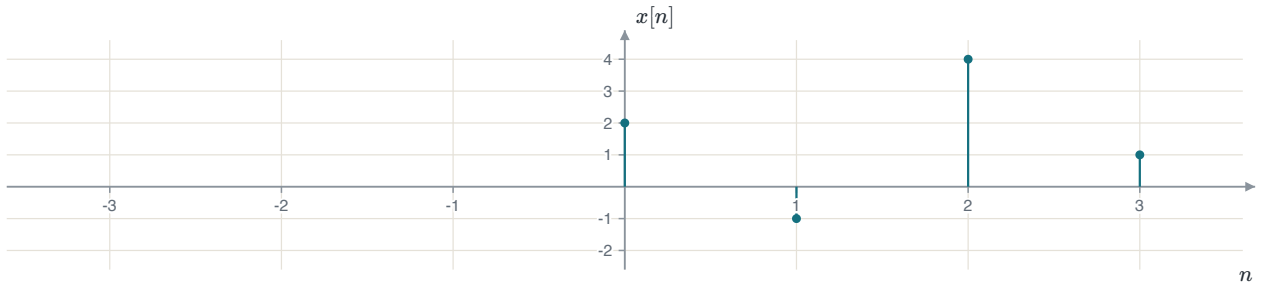
D1-13

Let $x(t) = 3e^{-4t}u(t)$.

- Find $\mathcal{E}_V\{x(t)\}$ and $\mathcal{O}dd\{x(t)\}$ and plot both.
- Calculate E_∞ for $x(t)$, for its even part, and for its odd part.
- Comment on the relation between the three energies.

D1-14

The sequence $x[n]$ equals 2, -1, 4, 1 at $n = 0, 1, 2, 3$ and is zero elsewhere.



- Plot $x[-n]$.
- Plot $\mathcal{E}_v\{x[n]\}$ and $\mathcal{O}_{dd}\{x[n]\}$.
- Verify that the two parts add back to $x[n]$ at $n = -1, n = 0$ and $n = 2$.

D1-15

Let

$$x(t) = \begin{cases} -1, & -2 \leq t < 0, \\ 2, & 0 \leq t \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

- Determine $x(-t)$ as a piecewise formula.
- Determine $\mathcal{E}_v\{x(t)\}$ and $\mathcal{O}_{dd}\{x(t)\}$, each as a piecewise formula.
- Calculate E_x , $E_{\mathcal{E}_v}$ and $E_{\mathcal{O}_{dd}}$, and verify that the two parts' energies add to E_x .

D1-16

Let $x[n]$ be an arbitrary real-valued sequence, not given by any formula.

- Prove that $\text{Odd}\{x\}[0] = 0$, directly from the definition of the odd part.
- The sequence $w[n]$ has $w[-1] = -5$, $w[0] = 0$, $w[1] = 5$, zero elsewhere. A student claims $\text{Odd}\{w\}[n] = w[n]$ for every n and $\mathcal{E}\mathcal{V}\{w\}[n] = 0$ for every n . Determine whether the claim is correct.
- A second sequence has $y[0] = 4$. Using part (a) only, and without computing $\text{Odd}\{y\}$, explain why $y[n]$ cannot be an odd sequence.

D1-17

Evaluate each of the following.

- $\int_{-\infty}^{\infty} (3t^2 + 1) [\delta(t + 2) + \delta(t - 4)] dt$
- $\int_{-\infty}^{\infty} e^{-2t} \sin(\pi t) \delta(t - 1.5) dt$
- $\int_{-\infty}^{\infty} t^2 \delta(4t - 8) dt$

- Evaluate the three integrals.
- State which property of the impulse each one uses, and say why the answer is a number rather than a signal.

D1-18

Let

$$g(t) = 3\delta(t + 1) - 2\delta(t - 2), \quad v(t) = \int_{-\infty}^t g(\tau) d\tau.$$

- Using $u(t) = \int_{-\infty}^t \delta(\tau) d\tau$, express $v(t)$ as a combination of unit step functions.
- Sketch $v(t)$ for $-3 \leq t \leq 4$.
- Read $v(0)$ and $v(3)$ from the sketch, and check them against the formula from part (a).

D1-19

Let $x[n]$ have $x[-1] = 5$, $x[0] = -3$, $x[1] = 2$, $x[2] = 7$, zero elsewhere.

- a) Write the sampling-property result $x[n]\delta[n-1]$ explicitly as a sequence.
- b) Write the sifting-property result $\sum_{n=-\infty}^{\infty} x[n]\delta[n-1]$ as a number, and confirm it equals $x[1]$.
- c) Using the representation property $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$, write out its four non-zero terms for this $x[n]$, and confirm the sum reproduces $x[0]$.

D1-20

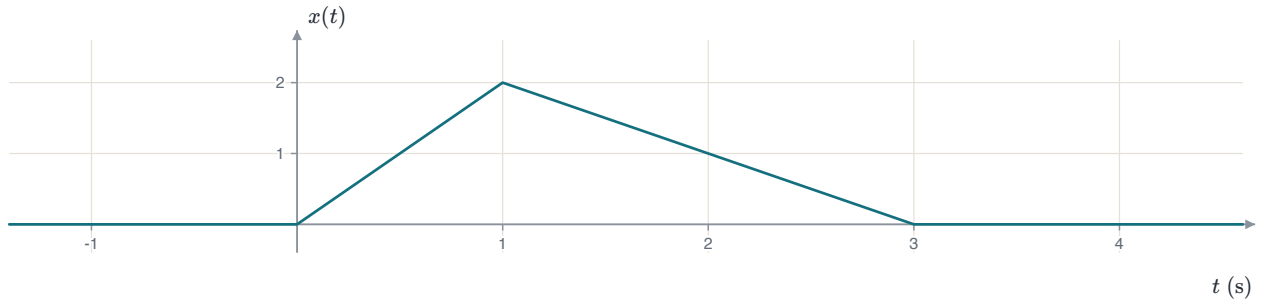
Let

$$x[n] = \delta[n] + 2\delta[n-2] - \delta[n-4], \quad y[n] = \sum_{k=-\infty}^n x[k].$$

- a) Evaluate $y[n]$ for $-2 \leq n \leq 6$.
- b) Express $y[n]$ in closed form as a combination of unit step functions, and check it against part (a).
- c) A continuous-time signal $p(t) = \delta(t) + 2\delta(t-2) - \delta(t-4)$ carries the same weights at the same locations. State its running integral $q(t) = \int_{-\infty}^t p(\tau) d\tau$ directly by analogy with $y[n]$, without repeating the calculation.

D1-21

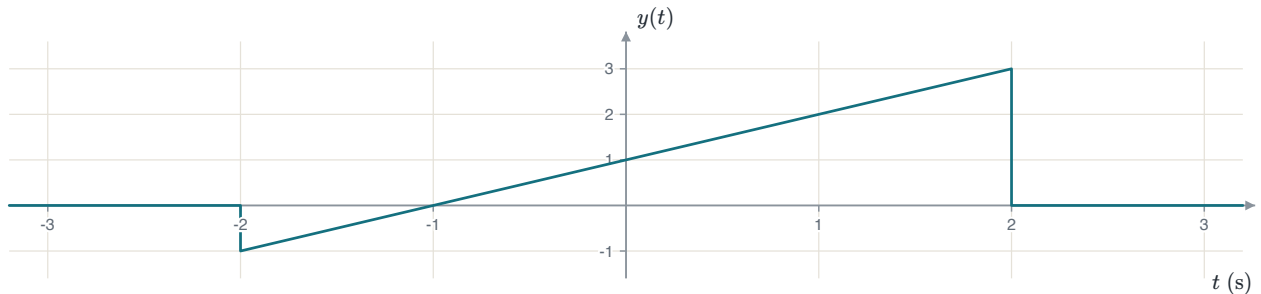
Work the following parts in order. Parts (b) to (e) all use the signal $x(t)$ plotted below.



- Is $x[n] = \cos\left(\frac{4\pi}{9}n - 2\right)$ periodic? If so, find its fundamental period.
- Calculate the energy E_∞ of $x(t)$.
- Plot $y(t) = x\left(3 - \frac{t}{2}\right)$.
- Plot $z(t) = \mathcal{E}\{y(t)\}$, the even part of the $y(t)$ signal of part (c).
- Evaluate $\int_{-\infty}^{\infty} z(t)\{\delta(t+4) + \delta(t-8)\} dt$, with $z(t)$ as defined in part (d).

D1-22

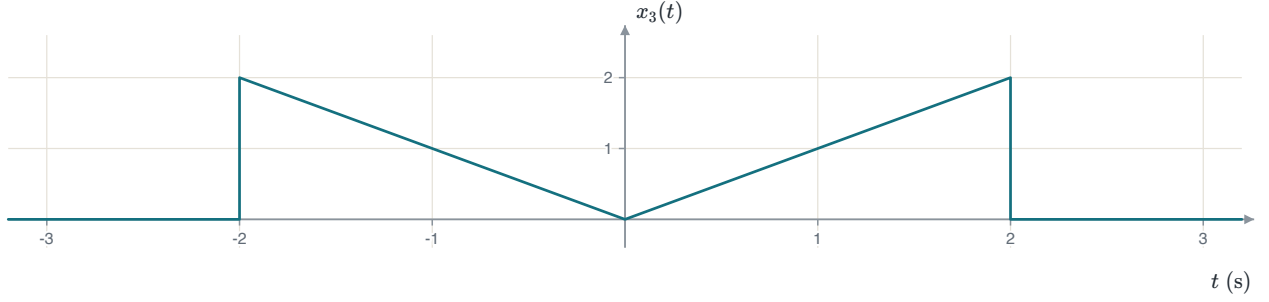
Work the following parts in order. Parts (c) and (d) use the signal $y(t)$ plotted below.



- Is the signal $x(t) = 2je^{j5t}$ periodic? If so, what is its fundamental period?
- Determine the energy E_∞ of the signal $x[n] = 3^{-n}u[n]$.
- Generate a plot of $y(-3t + 2)$ using the given signal $y(t)$.
- Plot the even part of $y(t)$.
- Evaluate $\int_{-\infty}^{\infty} e^{-t}\delta(2t - 4) dt$.

D1-23

Work the following parts in order. Parts (b) and (c) use the sequence $x_1[n]$ of part (a); parts (d) and (e) use the signal $x_3(t)$ plotted below.



- Sketch and label the signal $x_1[n] = \sum_{k=-\infty}^{\infty} \{\delta[n - 3k] - \delta[n + 1 + 4k]\}$.
- Is the $x_1[n]$ of part (a) periodic? If so, what is its fundamental period?
- For the $x_1[n]$ of part (a), plot the odd part of $x_2[n] = \begin{cases} x_1[n], & -3 \leq n \leq 3 \\ 0, & \text{otherwise.} \end{cases}$
- Calculate the energy E_{∞} of $x_3(t)$.
- For the $x_3(t)$ of part (d), evaluate $\int_{-\infty}^{\infty} x_3(t) \delta(t + 0.6) dt$.

D1-24

Consider the signal

$$x(t) = [t + 3\{u(t + 2) - u(t - 2)\}] \times \{u(t + 3) - u(t - 3)\}.$$

- Plot the even part of $x(t)$.
- Plot the odd part of $x(t)$.
- Calculate the energies of the even and odd parts of $x(t)$, and of $x(t)$ itself.

D1-25

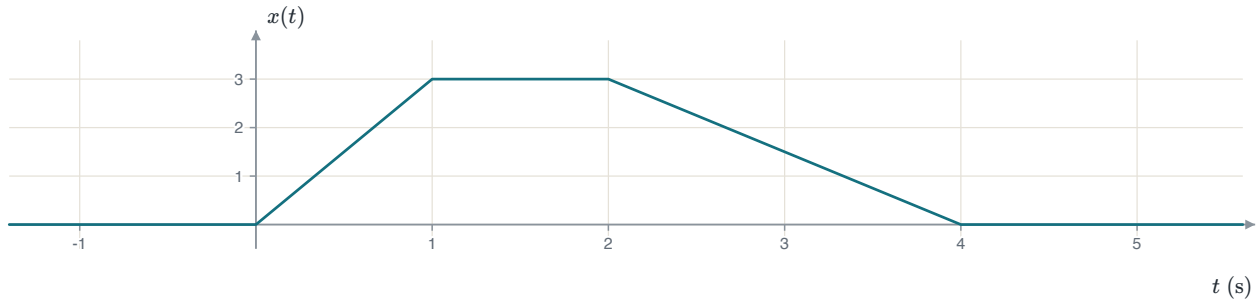
Two discrete-time complex exponentials differ only in whether π appears in the frequency:

$$x_1[n] = e^{j\frac{4}{7}n}, \quad x_2[n] = e^{j\frac{4\pi}{7}n}.$$

- Determine whether $x_1[n]$ is periodic. If it is, find its fundamental period.
- Determine whether $x_2[n]$ is periodic. If it is, find its fundamental period, and say what makes the two cases differ.

D1-26

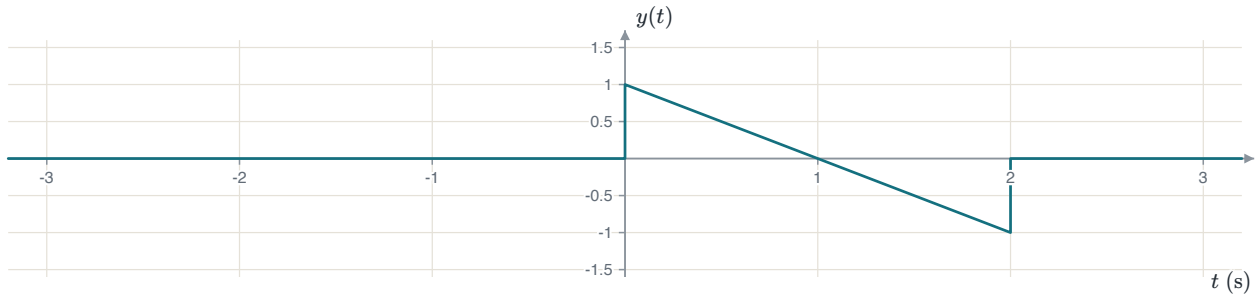
Work the following parts in order. Parts (b) to (e) all use the signal $x(t)$ plotted below.



- Is $x[n] = \sin\left(\frac{6\pi}{7}n + \frac{\pi}{4}\right)$ periodic? If so, find its fundamental period.
- Calculate the energy E_∞ of $x(t)$.
- Plot $y(t) = x(2t + 4)$.
- Plot $z(t) = \text{Odd}\{y(t)\}$, the odd part of the $y(t)$ signal of part (c).
- Evaluate $\int_{-\infty}^{\infty} z(t)\delta(3t + 3) dt$, with $z(t)$ as defined in part (d).

D1-27

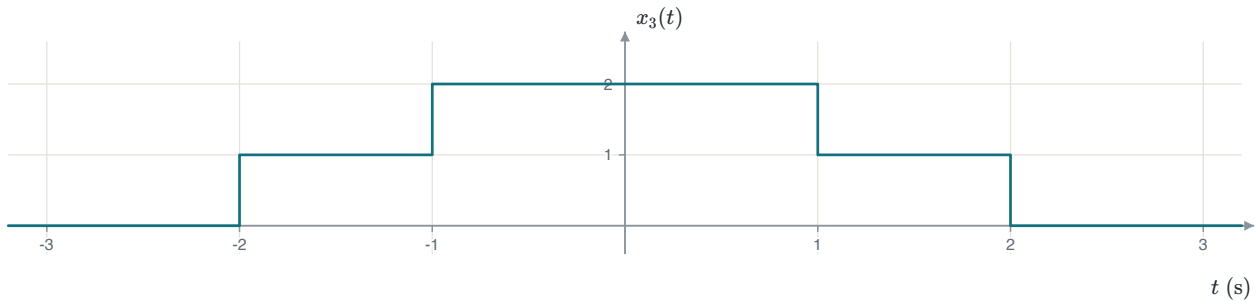
Work the following parts in order. Parts (c) to (e) use the signal $y(t)$ plotted below.



- Is the signal $x(t) = 5e^{j\frac{3\pi}{4}t}$ periodic? If so, what is its fundamental period?
- Determine the energy E_∞ of the signal $x[n] = \left(\frac{1}{2}\right)^n u[n - 2]$.
- Generate a plot of $y(2t - 1)$ using the given signal $y(t)$.
- Plot the even part of $y(t)$.
- Evaluate $\int_{-\infty}^{\infty} y(t)\delta(2t - 1) dt$.

D1-28

Work the following parts in order. Parts (b) and (c) use the sequence $x_1[n]$ of part (a); parts (d) and (e) use the signal $x_3(t)$ plotted below.



- Sketch and label the signal $x_1[n] = \sum_{k=-\infty}^{\infty} \{\delta[n - 4k] - \delta[n + 1 + 6k]\}$.
- Is the $x_1[n]$ of part (a) periodic? If so, what is its fundamental period?
- For the $x_1[n]$ of part (a), plot the odd part of $x_2[n] = \begin{cases} x_1[n], & -4 \leq n \leq 4 \\ 0, & \text{otherwise.} \end{cases}$
- Calculate the energy E_∞ of $x_3(t)$.
- For the $x_3(t)$ of part (d), evaluate $\int_{-\infty}^{\infty} x_3(t)\delta(t - 1.5) dt$.

D1-29

Consider the signal

$$x(t) = [2t + \{u(t+1) - u(t-1)\}] \times \{u(t+2) - u(t-2)\}.$$

- a) Plot the even part of $x(t)$.
- b) Plot the odd part of $x(t)$.
- c) Calculate the energies of the even and odd parts of $x(t)$, and of $x(t)$ itself.

D1-30

A signal is built from three step functions:

$$x(t) = u(t+2) - 2u(t) + u(t-2).$$

- a) Sketch $x(t)$ and give its piecewise form.
- b) Determine $\frac{dx}{dt}$ as a sum of impulses.
- c) Evaluate $\int_{-\infty}^{\infty} x(t)\delta(t-1) dt$.
- d) Evaluate $\int_{-\infty}^{\infty} \frac{dx}{dt} t^2 dt$.
- e) Calculate the energy E_{∞} of $x(t)$.

MODULE

2

Systems and Their Properties

30 questions on systems and their properties. Write your reasoning in the space under each one.

D2-01

Consider the system

$$y(t) = \cos(x(t)).$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer. An unjustified answer receives no credit.

D2-02

Consider the system

$$y[n] = x[n] x[n + 1].$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-03

Consider the system

$$y(t) = x^3(t) - x(t).$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-04

Two memoryless systems:

$$(i) \ y(t) = x^3(t), \quad (ii) \ z[n] = x[n] x[n-2].$$

- a) Determine whether system (i) is invertible. Give the inverse relation if it is.
- b) Determine whether system (ii) is invertible. Give an explicit counterexample if it is not.
- c) Explain why raising the input to an odd power preserves invertibility while forming a product of two different samples generally does not.

D2-05

Consider the system

$$y[n] = \sum_{k=n-3}^n x[k].$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-06

Consider the system

$$y(t) = \int_0^t x(\tau) d\tau.$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-07

Consider the system

$$y[n] = \sum_{k=-\infty}^n \left(\frac{1}{2}\right)^{n-k} x[k].$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
 b) Justify each answer.

D2-08

The running sum

$$y[n] = \sum_{k=-\infty}^n x[k]$$

is unstable in general. A particular input is

$$x[n] = \left(\frac{1}{3}\right)^n u[n].$$

- a) Verify that $x[n]$ is bounded, and find a closed form for $y[n]$ valid for $n \geq 0$.
 b) Show that this particular response is itself a bounded sequence.
 c) Explain why part (b) does not establish that the system is BIBO stable, and produce a single bounded input whose response is unbounded.
 d) State, in one sentence, the general distinction between a property of a system and a property of one of its responses.

D2-09

Consider the system

$$y(t) = e^{-t} x(t).$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
 b) Justify each answer.

D2-10

Consider the system

$$y[n] = 2^{-|n|}x[n].$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-11

Consider the system

$$y(t) = x(t) \cos(\pi t) + x(t - 1).$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-12

Two systems built from a time-varying gain:

$$(i) \ y(t) = (t^2 + 1)x(t), \quad (ii) \ y(t) = tx(t).$$

- a) Determine whether system (i) is invertible. Give the inverse relation if it is.
- b) Determine whether system (ii) is invertible. Give an explicit counterexample if it is not.
- c) State the general condition on a gain $g(t)$, in the relation $y(t) = g(t)x(t)$, for the system to be invertible.

D2-13

Consider the system

$$y[n] = x[n + 2].$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-14

Consider the system

$$y(t) = x(3 - t).$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-15

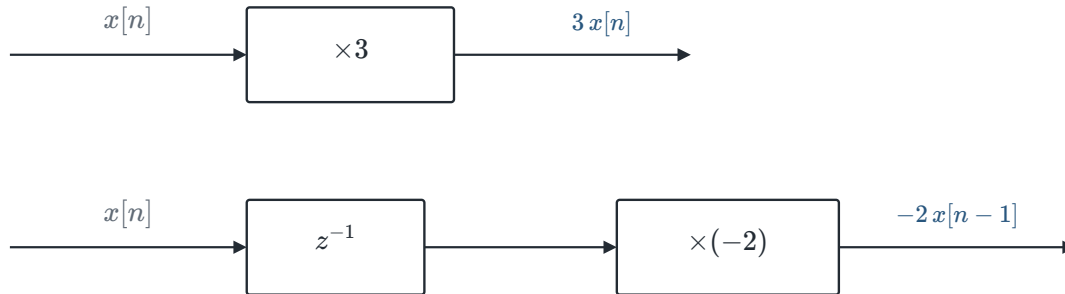
Consider the system

$$y[n] = x[3n - 1].$$

- a) Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- b) Justify each answer.

D2-16

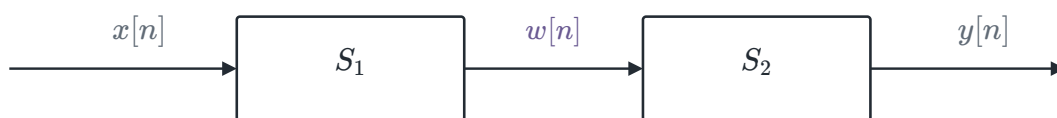
The block diagram below shows a system built from a one-sample delay and two fixed gains. The direct branch multiplies the input by 3; the delayed branch multiplies the once-delayed input by -2 ; the two branches are added.



- Read the diagram and write the input–output relation it implements.
- Determine whether the system is memoryless, linear, time invariant, causal and BIBO stable.
- State the condition for the system to be invertible, and give the inverse relation.

D2-17

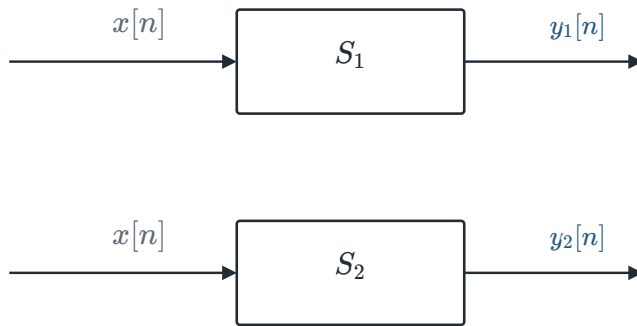
Two systems are connected in series, the output of S_1 feeding directly into S_2 . S_1 is the first difference, $w[n] = x[n] - x[n-1]$; S_2 is a fixed gain, $y[n] = 2w[n]$.



- Write the overall input–output relation of the series connection, $y[n]$ in terms of $x[n]$ alone.
- S_1 and S_2 are each linear and each time invariant. Argue directly from the definitions that a series connection of two linear systems is linear, and that a series connection of two time-invariant systems is time invariant.
- Verify both conclusions directly from the relation found in part (a).

D2-18

Two systems are connected in parallel, both acting on the same input and their outputs added: S_1 is the identity, $y_1[n] = x[n]$; S_2 is a unit delay, $y_2[n] = x[n - 1]$.



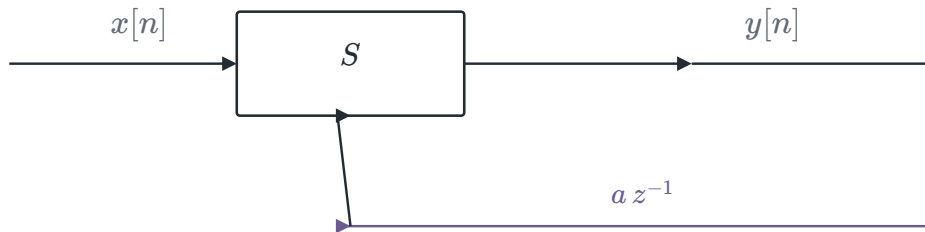
- a) Write the overall input–output relation of the parallel connection.
- b) S_1 and S_2 are each causal and each BIBO stable. Argue directly from the definitions that a parallel connection of two causal systems is causal, and that a parallel connection of two BIBO-stable systems is BIBO stable.
- c) Confirm both conclusions directly for this particular combination.

D2-19

The system below is built with feedback:

$$y[n] = x[n] - a y[n-1],$$

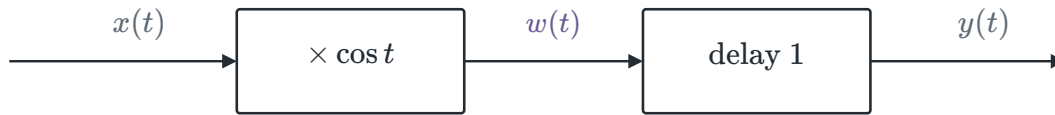
where a is a real constant, the output returning through a gain a and a one-sample delay and entering with a negative sign.



- Determine whether the system is memoryless, causal and linear, for any real value of a .
- Iterate the recursion to express $y[n]$ as a sum involving only $x[n], x[n-1], x[n-2], \dots$, assuming $y[k] \rightarrow 0$ as $k \rightarrow -\infty$.
- Using the sum from part (b), state the condition on a under which the system is BIBO stable, and justify it.

D2-20

Two systems are connected in series: S_1 is a modulator, $w(t) = x(t) \cos(t)$; S_2 is a fixed one-second delay, $y(t) = w(t - 1)$.



- Determine whether S_1 , taken alone, is time invariant.
- Determine whether S_2 , taken alone as a system with input w and output y , is time invariant.
- Write the overall relation for the series connection, and determine whether it is time invariant. Justify with a counterexample if it fails.
- Determine whether the overall connection is linear, using the fact that a series connection of two linear systems is linear.

D2-21

Consider the following input and output relationship of a system

$$y[n] = \sum_{k=0}^{n+2} \sin\left(\frac{\pi}{6}k\right) x[k].$$

- Determine whether the system is memoryless.
- Determine whether the system is linear.
- Determine whether the system is time-invariant.
- Determine whether the system is causal.
- Determine whether the system is stable. Justify every answer; an unjustified answer earns nothing.

D2-22

Consider the following input-output relationship of a system.

$$y(t) = (t + 2)x(t - 3)$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-23

Consider the following input and output relationship of a system

$$y(t) = \text{Odd}\{x(t)\} = \frac{1}{2}[x(t) - x(-t)].$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-24

Consider the following input and output relationship of a system

$$y[n] = x[n]x[n - 2].$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-25

Consider the following input-output relationship of a system.

$$y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-26

Consider the following input and output relationship of a system

$$y[n] = n x[n + 1].$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-27

Consider the following input-output relationship of a system.

$$y(t) = e^{x(t)}$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-28

Consider the following input and output relationship of a system

$$y[n] = \sum_{k=n-3}^{n+3} x[k].$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-29

Consider the following input-output relationship of a system.

$$y(t) = x\left(\frac{t}{3}\right)$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

D2-30

Consider the following input and output relationship of a system

$$y(t) = x(t) \cos(3t).$$

- a) Determine whether the system is memoryless.
- b) Determine whether the system is linear.
- c) Determine whether the system is time-invariant.
- d) Determine whether the system is causal.
- e) Determine whether the system is stable. Justify every answer.

MODULE

3

Linear Time-Invariant Systems

30 questions on linear time-invariant systems. Write your reasoning in the space under each one.

D3-01

The input and output of a discrete-time LTI system are related by

$$y[n] = 2x[n] - x[n-1] + 3x[n-3].$$

- a) Determine and plot the impulse response $h[n]$.
- b) For the input $x[n] = u[n] - u[n-3]$, compute and plot $y[n] = x[n] * h[n]$.

D3-02

A discrete-time LTI system is described by

$$y[n] = \frac{1}{4} y[n-1] + x[n],$$

and is initially at rest.

- a) Determine and plot the impulse response $h[n]$.
- b) For the input $x[n] = u[n]$, compute and plot $y[n]$.
- c) State the limit of $y[n]$ as $n \rightarrow \infty$ and say what it means.

D3-03

A discrete-time LTI system is described by

$$y[n] = x[n+1] - 2x[n] + x[n-1],$$

which uses a future value of the input and is therefore not causal.

- a) Determine and plot the impulse response $h[n]$.
- b) For the input $x[n] = u[n+1] - u[n-2]$ (equal to 1 at $n = -1, 0, 1$), compute and plot $y[n] = x[n] * h[n]$.
- c) Using only $\sum_n h[n]$ and $\sum_n x[n]$, state $\sum_n y[n]$ without adding the samples of part (b), and confirm the two agree.

D3-04

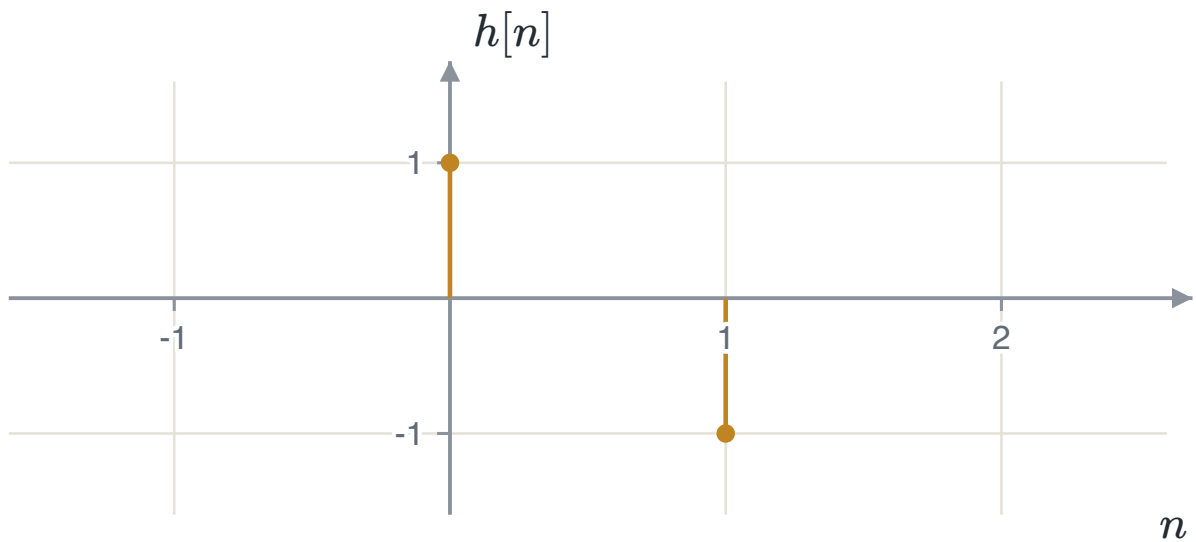
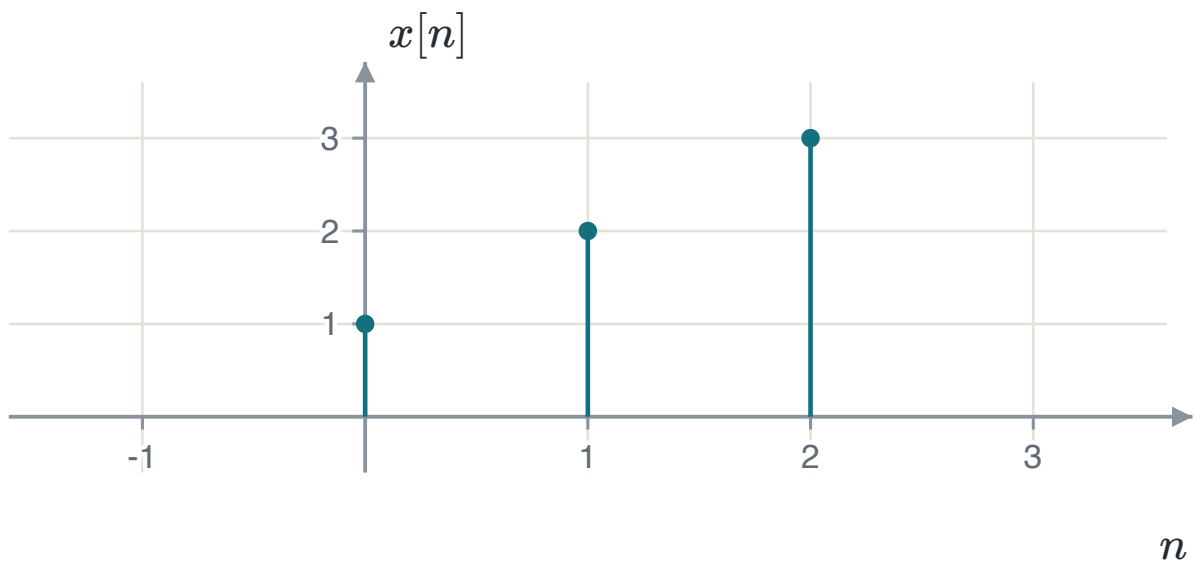
Two discrete-time LTI systems are described by their own difference equations,

$$y_1[n] = x[n] + x[n-1] \quad \text{and} \quad y_2[n] = x[n] - x[n-1].$$

- a) Determine $h_1[n]$ and $h_2[n]$.
- b) The two systems are placed in cascade. Determine and plot the impulse response of the cascade, $h_c[n] = h_1[n] * h_2[n]$.
- c) The same two systems are placed in parallel instead, outputs summed. Determine $h_p[n] = h_1[n] + h_2[n]$, and use it to find the response to $x[n] = 3\delta[n-4]$.

D3-05

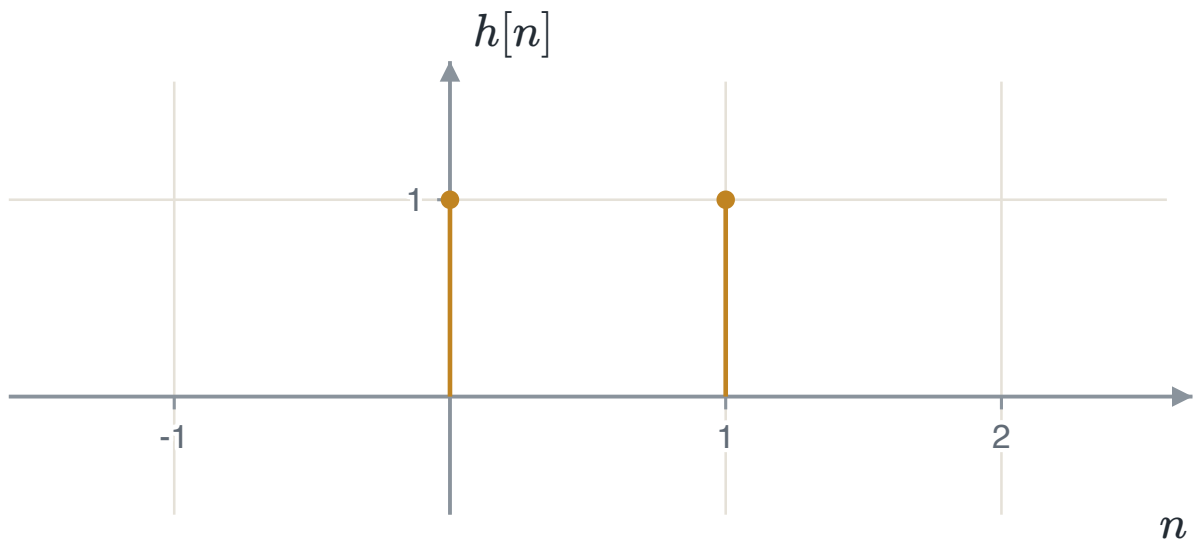
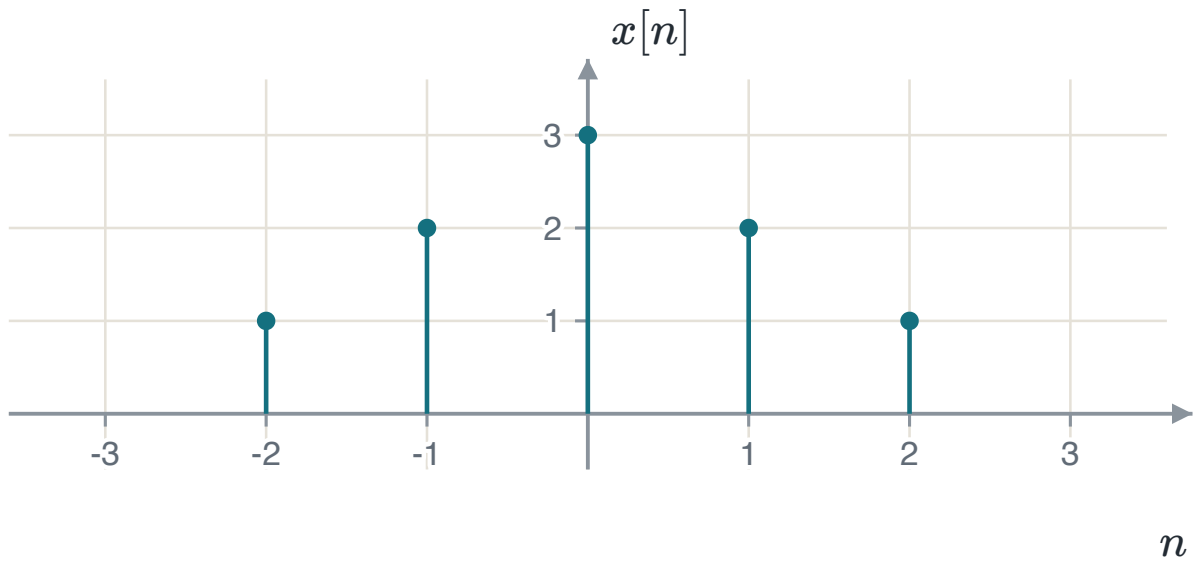
The sequences $x[n]$ and $h[n]$ of a discrete-time LTI system are shown below.



- Compute and plot $y[n] = x[n] * h[n]$.
- State which sample of $y[n]$ has the largest magnitude, and explain why.

D3-06

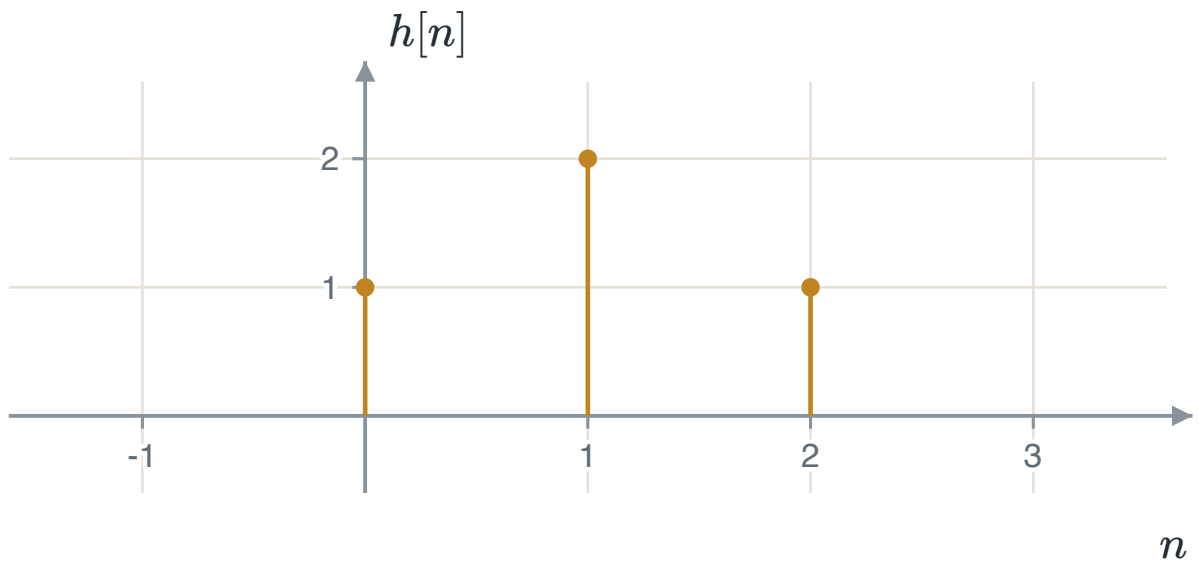
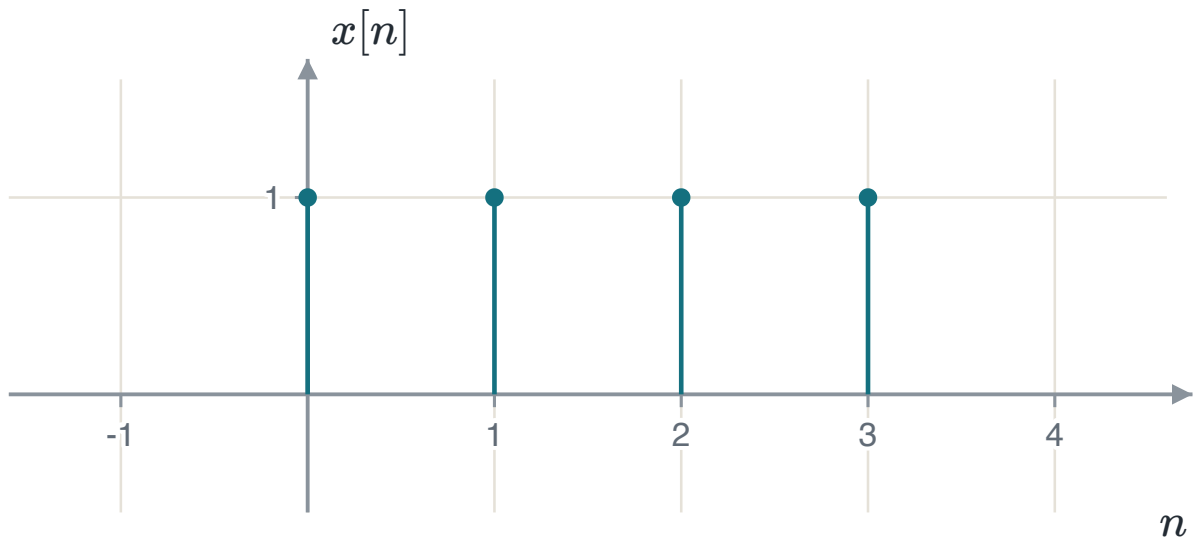
The sequences $x[n]$ and $h[n]$ of a discrete-time LTI system are shown below.



- Compute and plot $y[n] = x[n] * h[n]$.
- Verify that $y[n]$ is symmetric about $n = 0.5$, and say why that point is expected.

D3-07

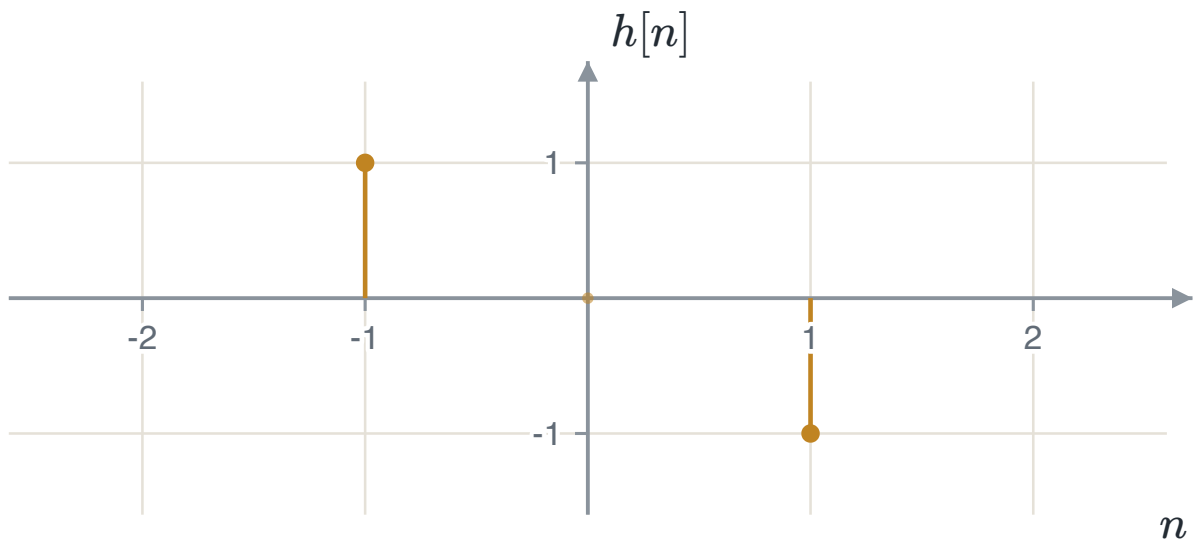
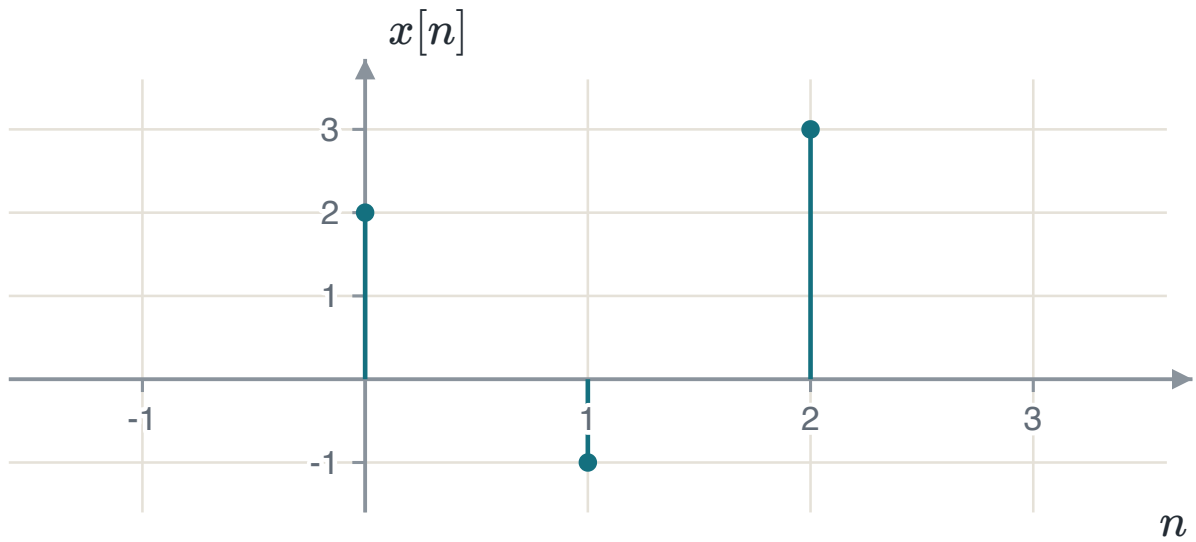
The sequences $x[n]$ and $h[n]$ of a discrete-time LTI system are shown below.



- Compute and plot $y[n] = x[n] * h[n]$.
- Confirm that the peak value of $y[n]$ equals $\sum_n h[n]$, and state the two indices where the peak occurs.

D3-08

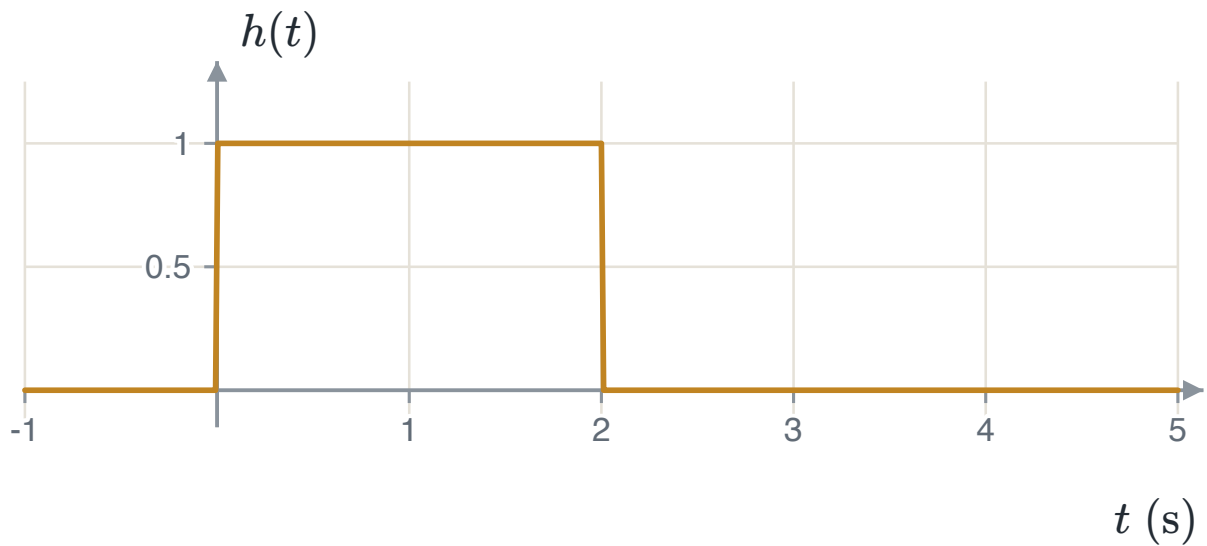
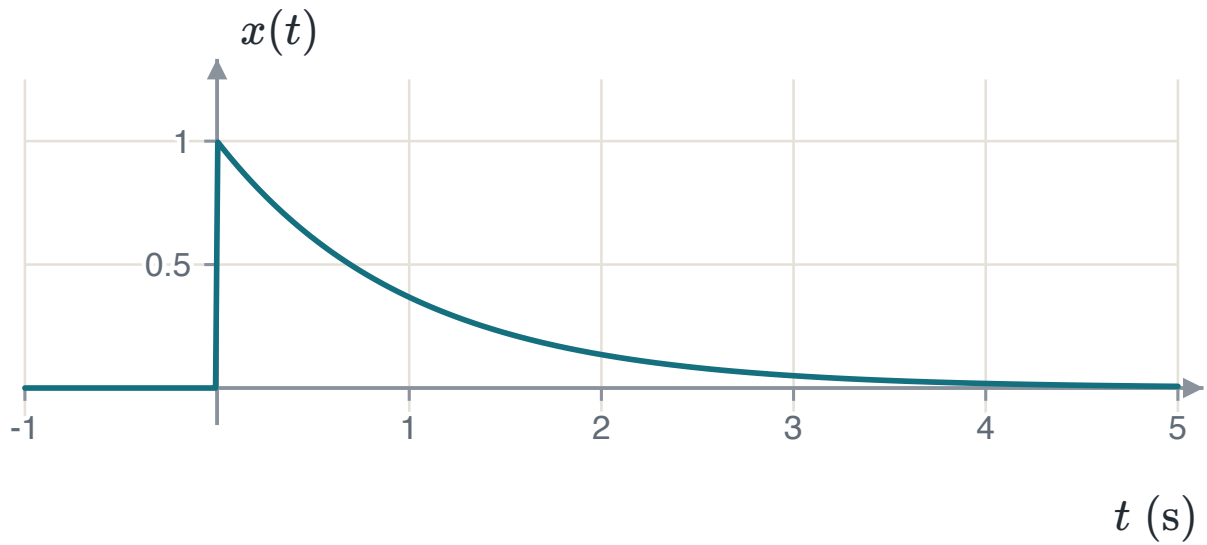
The sequences $x[n]$ and $h[n]$ of a discrete-time LTI system are shown below.



- Compute and plot $y[n] = x[n] * h[n]$.
- Show that $y[n] = x[n+1] - x[n-1]$ directly from the two non-zero locations of $h[n]$, and use it to check one sample of part (a) without repeating the full sum.

D3-09

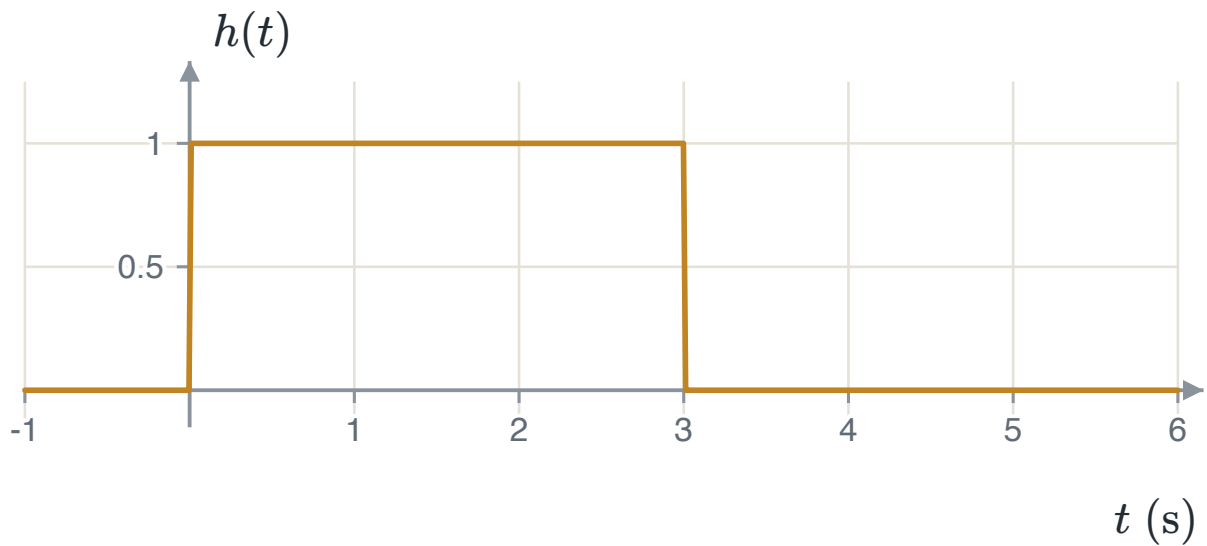
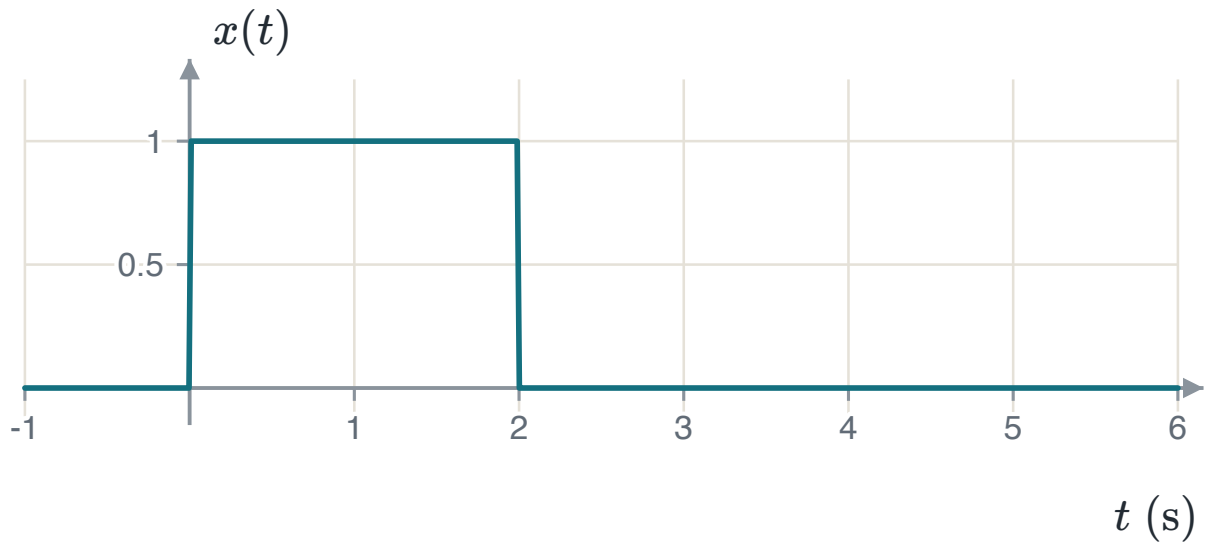
Let $x(t) = e^{-t}u(t)$ and $h(t) = u(t) - u(t - 2)$.



- Compute $y(t) = x(t) * h(t)$ for $0 \leq t < 2$.
- Compute $y(t)$ for $t \geq 2$.
- Confirm the two branches agree at $t = 2$, and state $\lim_{t \rightarrow \infty} y(t)$.

D3-10

Let $x(t) = 1$ for $0 < t < 2$ and $h(t) = 1$ for $0 < t < 3$, both zero elsewhere.



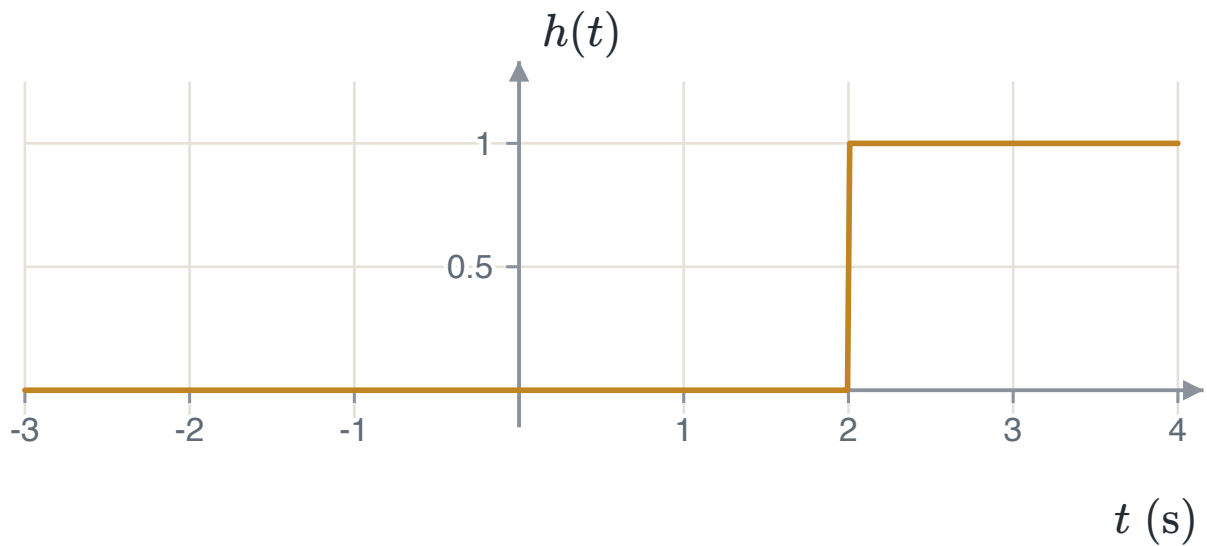
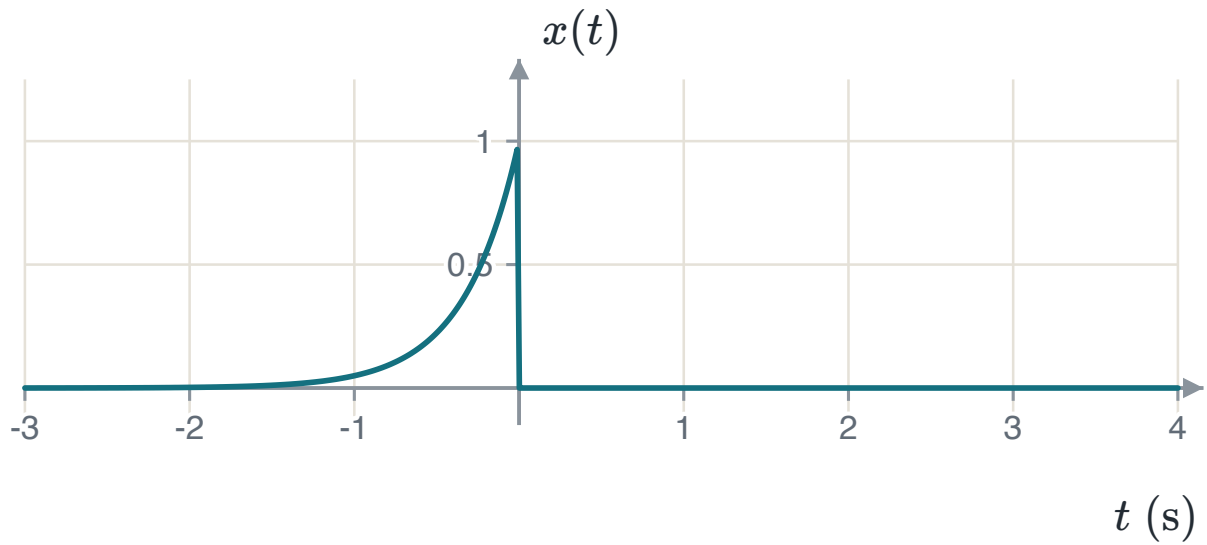
- List the four case boundaries and say where each comes from.
- Compute $y(t) = x(t) * h(t)$ in each of the three non-zero cases.
- Verify that the total area under $y(t)$ equals the product of the two pulse widths.

D3-11

Let $x(t) = e^{-2t}u(t)$ and $h(t) = e^{-3t}u(t)$.

- a) Compute $y(t) = x(t) * h(t)$ for $t \geq 0$, stating the case for $t < 0$ separately.
- b) Find the time t^* at which $y(t)$ is maximum, and the peak value.
- c) Evaluate $y(0)$ directly from the convolution integral, without using the closed form of part (a).

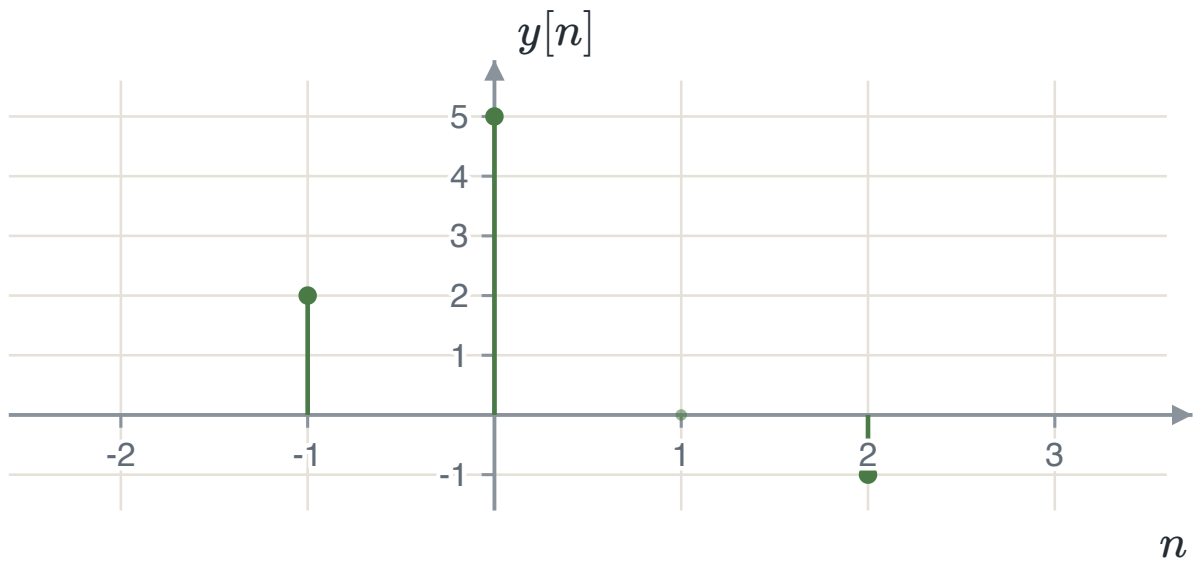
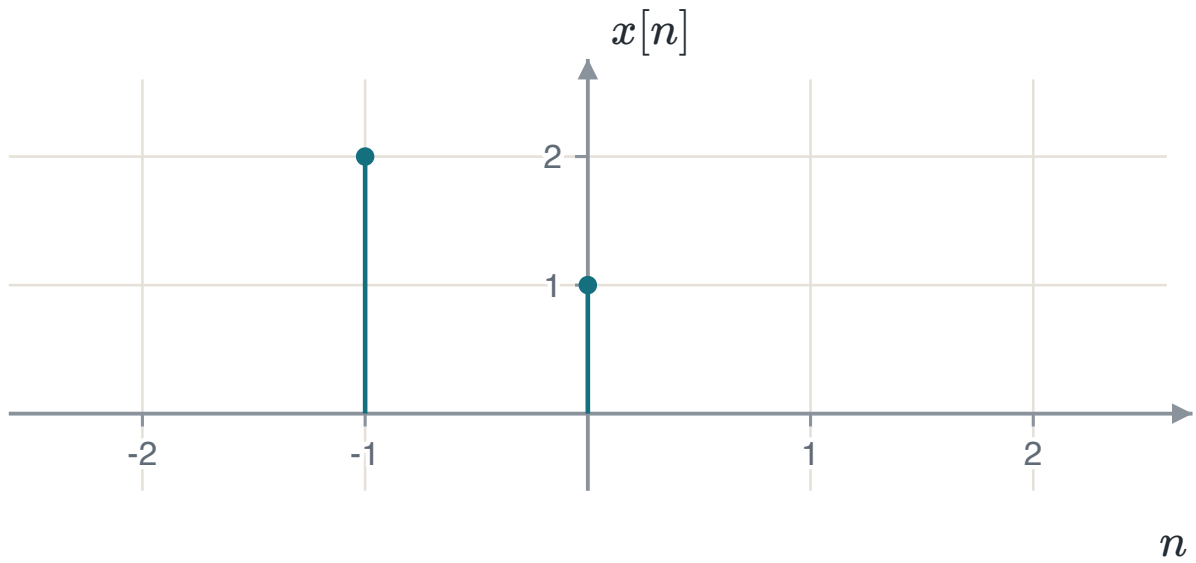
D3-12

 Let $x(t) = e^{3t}u(-t)$ and $h(t) = u(t - 2)$.


- Compute $y(t) = x(t) * h(t)$ for $t < 2$.
- Compute $y(t)$ for $t \geq 2$.
- Confirm the two branches agree at $t = 2$, and interpret the system as a delayed accumulator.

D3-13

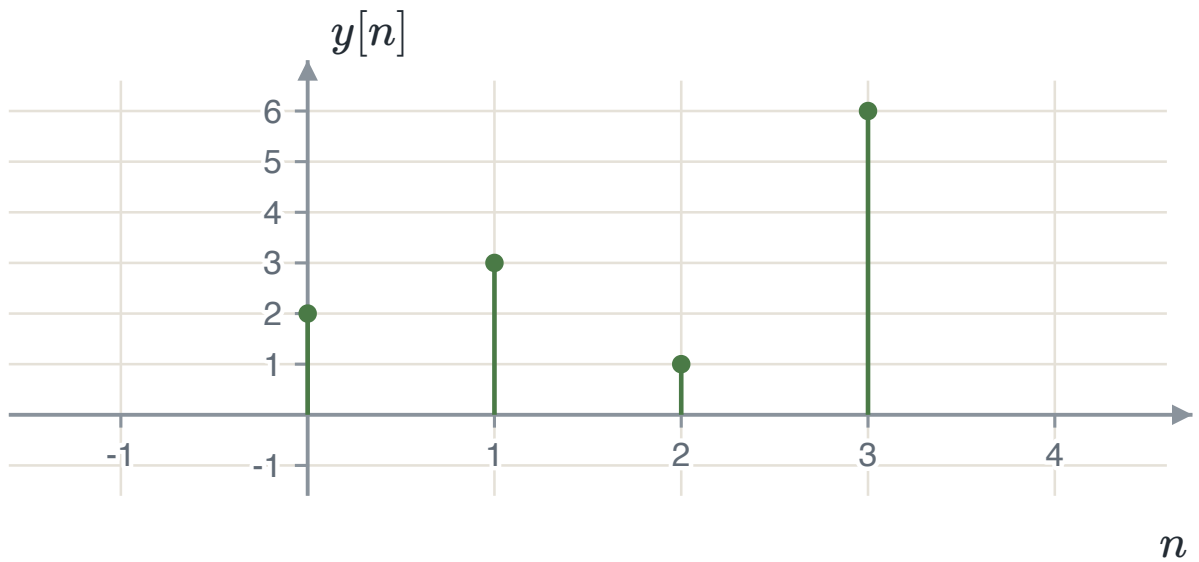
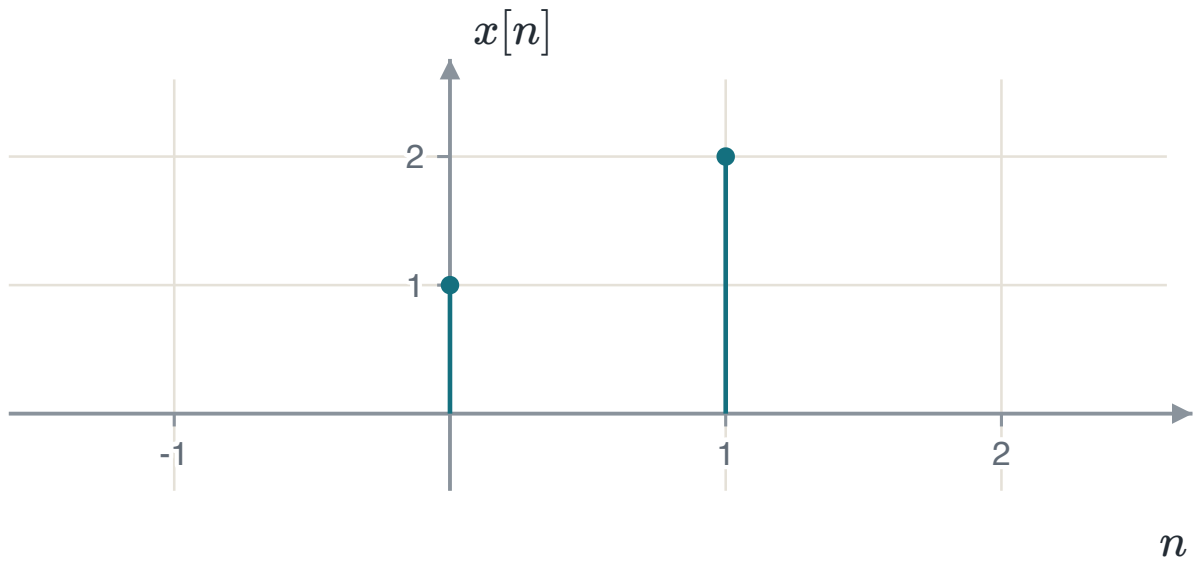
A discrete-time LTI system has $x[n] = 2\delta[n + 1] + \delta[n]$ at its input. The measured output $y[n]$ is shown below. The system is known to be causal, with an impulse response at most three samples long.



- Using $y[-1]$ alone, find $h[0]$.
- Using $y[0]$ and $h[0]$, find $h[1]$; then using $y[1]$, find $h[2]$.
- Confirm $h[2]$ against $y[2]$, without introducing a new unknown.

D3-14

A discrete-time LTI system has $x[n] = \delta[n] + 2\delta[n - 1]$ at its input. The measured output $y[n]$ is shown below. The system is known to be causal, with an impulse response at most three samples long.



- Using $y[0]$ alone, find $h[0]$.
- Using $y[1]$ and $h[0]$, find $h[1]$; then using $y[2]$, find $h[2]$.
- Confirm $h[2]$ against $y[3]$, without introducing a new unknown.

D3–15

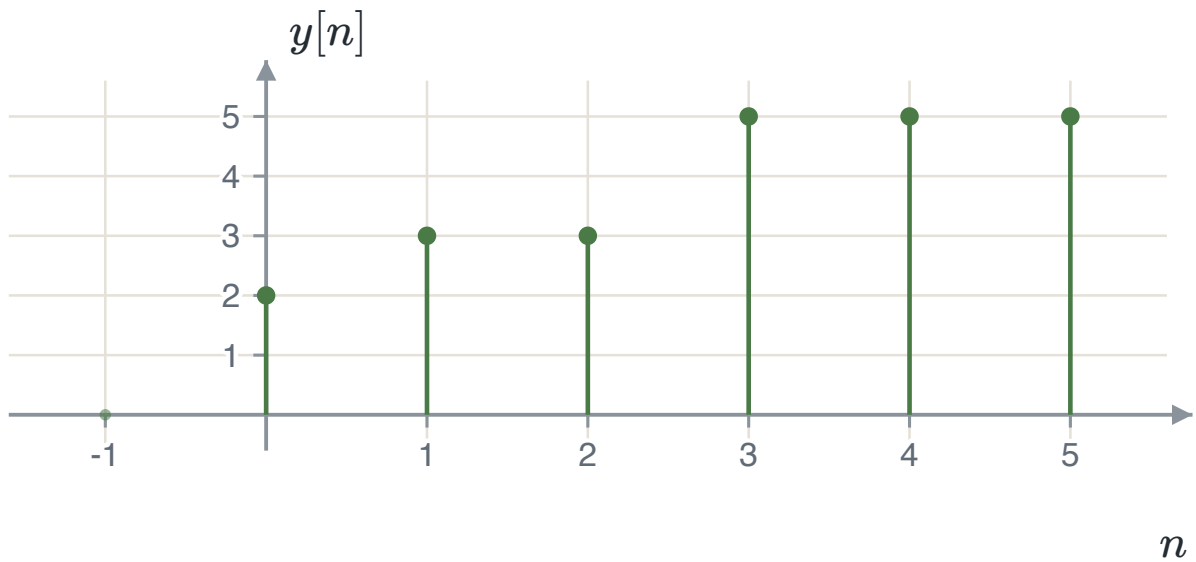
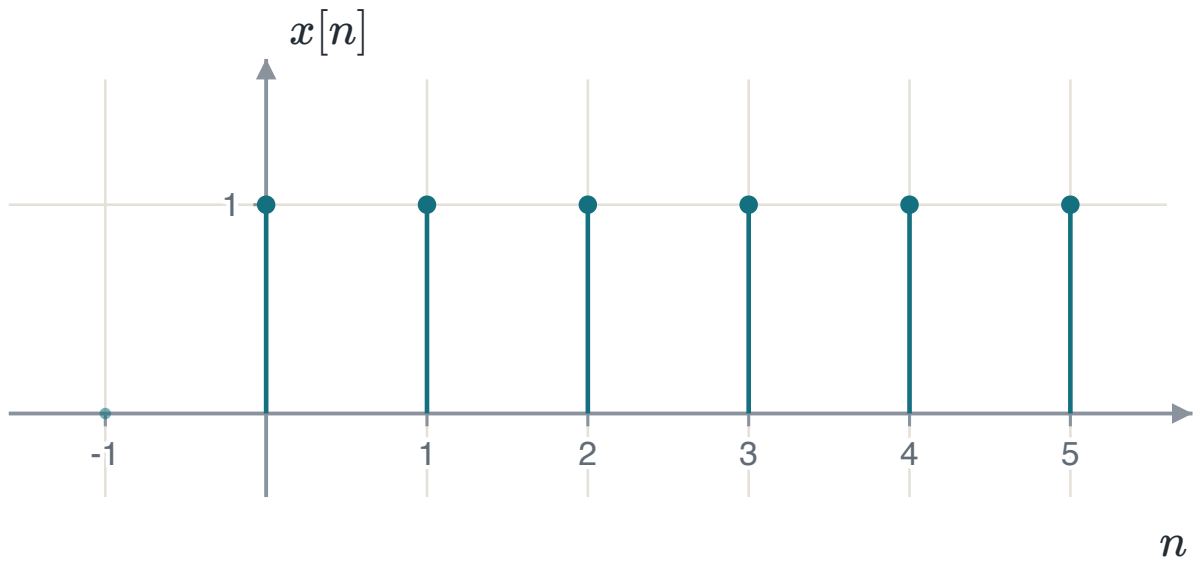
A discrete-time LTI system has impulse response $h[n] = \delta[n] - \frac{1}{2}\delta[n - 1]$, shown below.



- a) Find the impulse response $g[n]$ of the causal inverse system, defined by $h[n] * g[n] = \delta[n]$.
- b) Determine whether $g[n]$ is causal and whether it is stable.
- c) Verify $h[n] * g[n] = \delta[n]$ directly at $n = 0, 1, 2$.

D3-16

A discrete-time LTI system is driven by the unit step $x[n] = u[n]$. The measured output $y[n]$ is shown below.



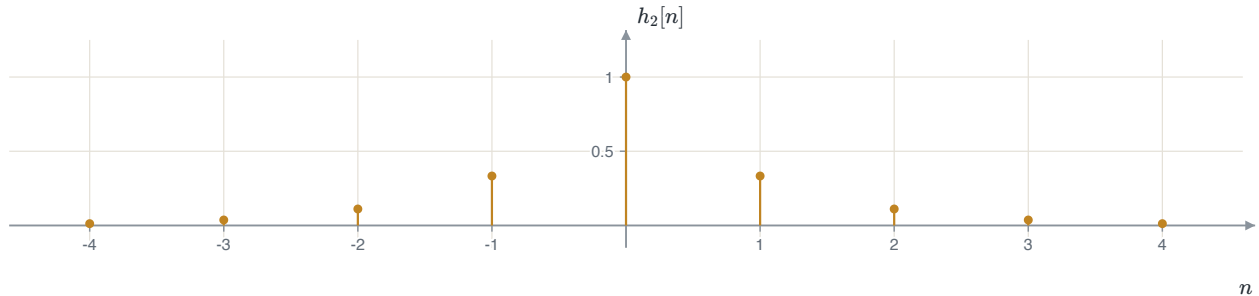
- Explain why $h[n] = y[n] - y[n-1]$ when the input is $x[n] = u[n]$.
- Use that relation to find $h[0]$, $h[1]$, $h[2]$, $h[3]$.
- State $\sum_n h[n]$ directly from the plotted $y[n]$, without adding the four values of part (b).

D3-17

Two discrete-time LTI systems have impulse responses

$$h_1[n] = 2^n u[n] \quad \text{and} \quad h_2[n] = \left(\frac{1}{3}\right)^{|n|}.$$

$h_2[n]$ is shown below; it is two-sided.



- Determine whether $h_1[n]$ is causal, and whether it is stable.
- Determine whether $h_2[n]$ is causal, and whether it is stable.
- Say what the pairing of these two results shows about the two properties.

D3-18

A continuous-time LTI system has impulse response $h(t) = e^{2t}u(-t)$.

- Determine whether the system is causal.
- Determine whether it is stable.
- For the bounded input $x(t) = u(t)$, compute $y(t) = x(t) * h(t)$ explicitly and confirm it stays within the bound predicted by stability.

D3-19

A discrete-time LTI system has impulse response $h[n] = (0.6)^n u[n]$.

- Find and plot the step response $s[n] = h[n] * u[n]$.
- Recover $h[n]$ from $s[n]$ using $h[n] = s[n] - s[n-1]$, and confirm it reproduces the given $h[n]$.
- State $\lim_{n \rightarrow \infty} s[n]$ and identify it as $\sum_n h[n]$.

D3-20

A continuous-time LTI system has impulse response $h(t) = e^{-4t}u(t)$.

- a) Find and plot the step response $s(t) = h(t) * u(t)$.
- b) Recover $h(t)$ from $s(t)$ using $h(t) = ds/dt$, and confirm it reproduces the given $h(t)$ for $t > 0$.
- c) State $\lim_{t \rightarrow \infty} s(t)$ and identify it as $\int h(t) dt$.

D3-21

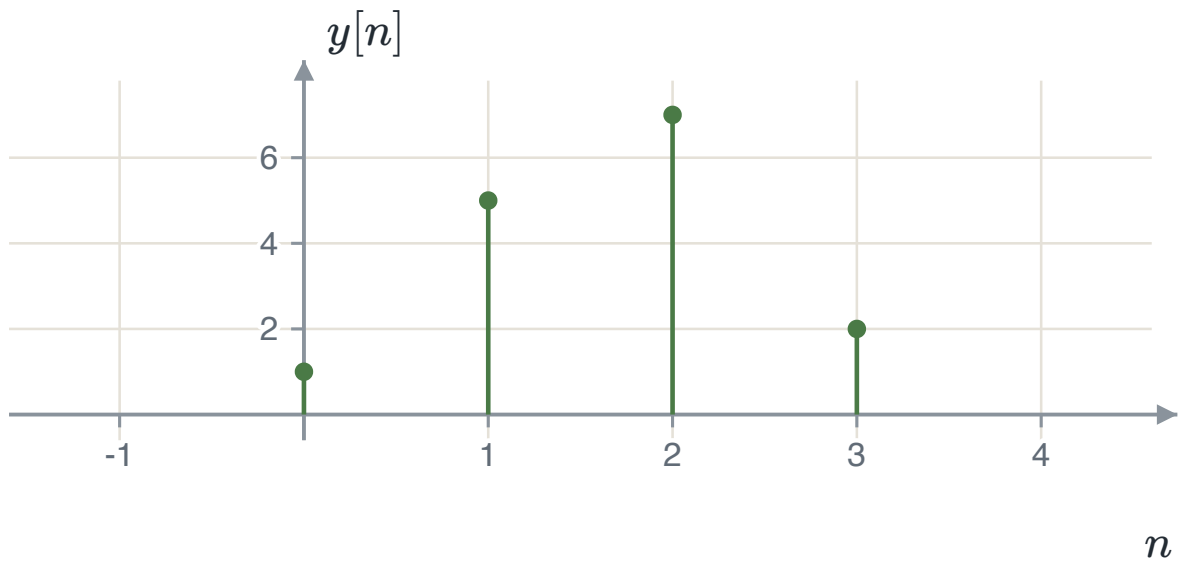
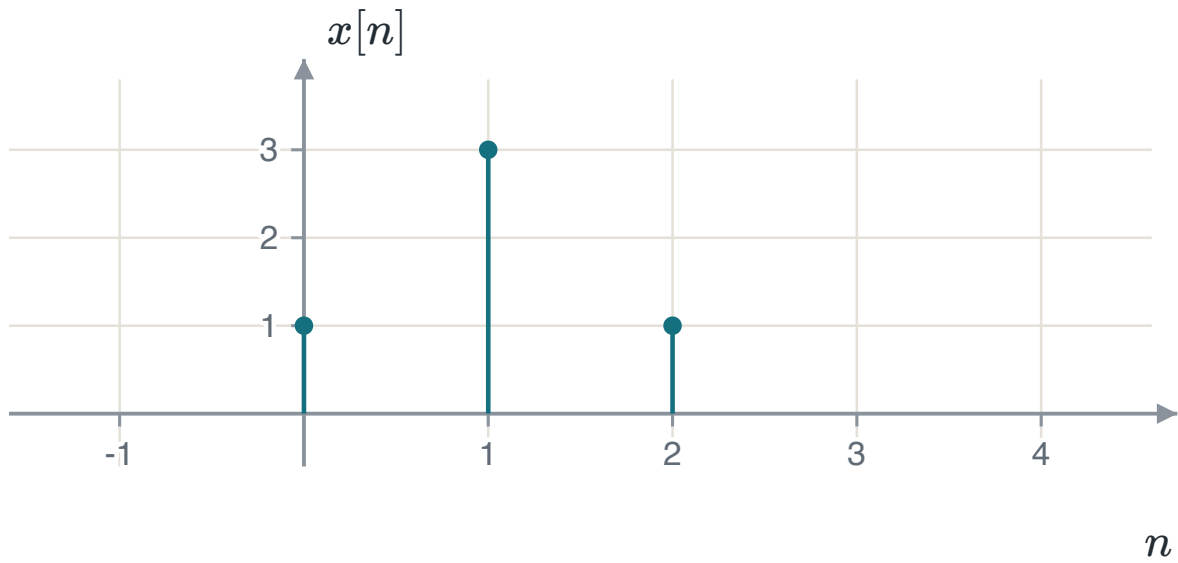
The input and output relationship of an LTI system is described as

$$y[n] = 3x[n] + x[n-2] + 2x[n-3].$$

- a) Determine and plot the impulse response of this LTI system, $h[n]$.
- b) For the input signal $x[n] = \sum_{k=-1}^2 k \delta[n-2k]$, compute and plot the output signal $y[n] = x[n] * h[n]$, using the discrete-time convolution.

D3-22

In the figure below, the signal $x[n]$ is input into a linear time-invariant (LTI) system, resulting in $y[n]$ as the output.



- a) Determine and plot the impulse response of this LTI system, $h[n]$. *Hint: attempt to express $y[n]$ as a function of $x[n]$.*
- b) Given the input signal $x_1[n] = 2\delta[n+1] - \delta[n] + \delta[n-2]$, sketch the output signal $y_1[n] = x_1[n] * h[n]$, using the discrete-time convolution.

D3-23

The input and output relationship of an LTI system is described as

$$y[n] = \frac{1}{3}y[n-1] + x[n].$$

- a) Determine and plot the impulse response of this LTI system, $h[n]$.
- b) For the input signal $x[n] = u[n]$, compute and plot the output signal $y[n] = x[n] * h[n]$, using the discrete-time convolution.

D3-24

Let $x(t) = e^{-2|t|}[u(t+1) - u(t-1)]$ be an input to the LTI system which has the following impulse response, $h(t) = u(t)$. Compute and plot the output signal $y(t) = x(t) * h(t)$, using the continuous-time convolution.

- a) Write the convolution integral and identify the ranges of t that need separate treatment.
- b) Evaluate the integral on each range and plot $y(t)$.

D3-25

Let

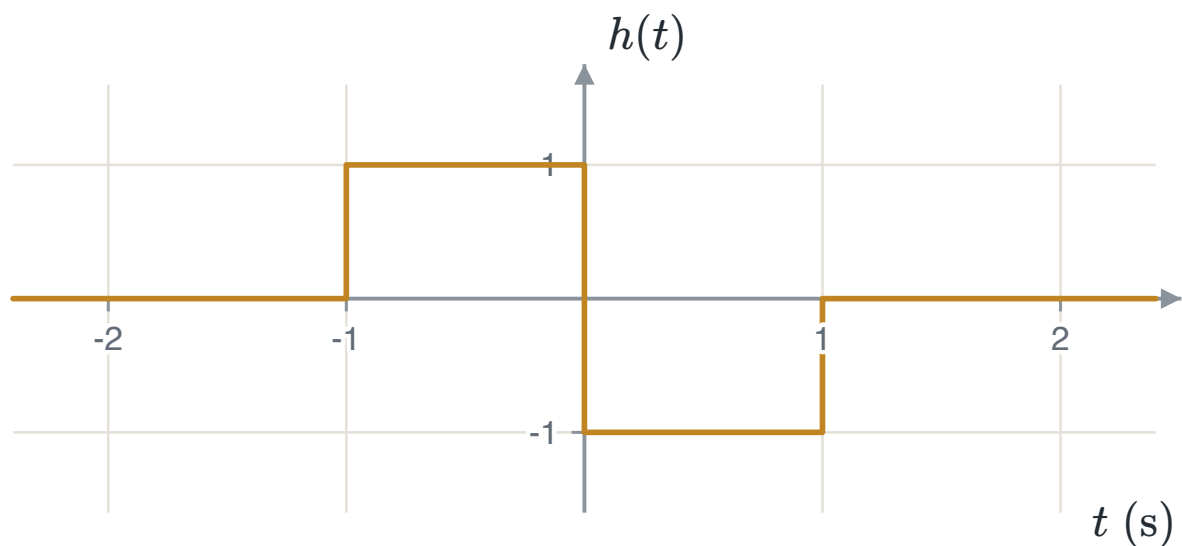
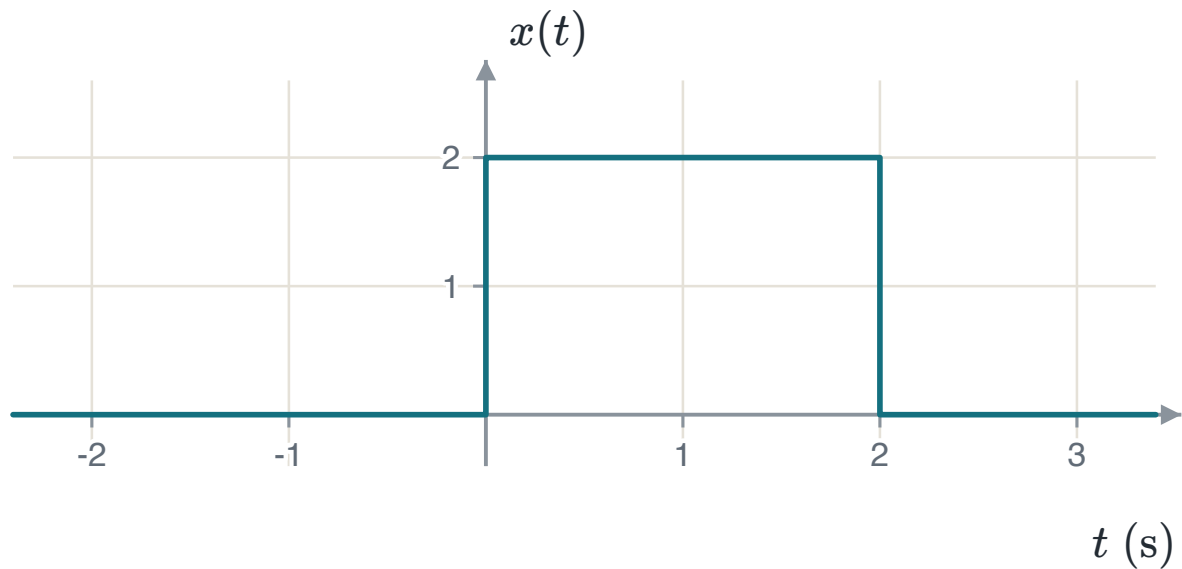
$$x(t) = \begin{cases} e^{-3t}, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

be an input to the LTI system which has the following impulse response, $h(t) = u(t) - u(t-2)$. Compute the output signal $y(t) = x(t) * h(t)$, using the continuous-time convolution.

- a) Identify the ranges of t over which the overlap of $x(\tau)$ and $h(t-\tau)$ changes.
- b) Evaluate the convolution integral on each range and plot $y(t)$.

D3-26

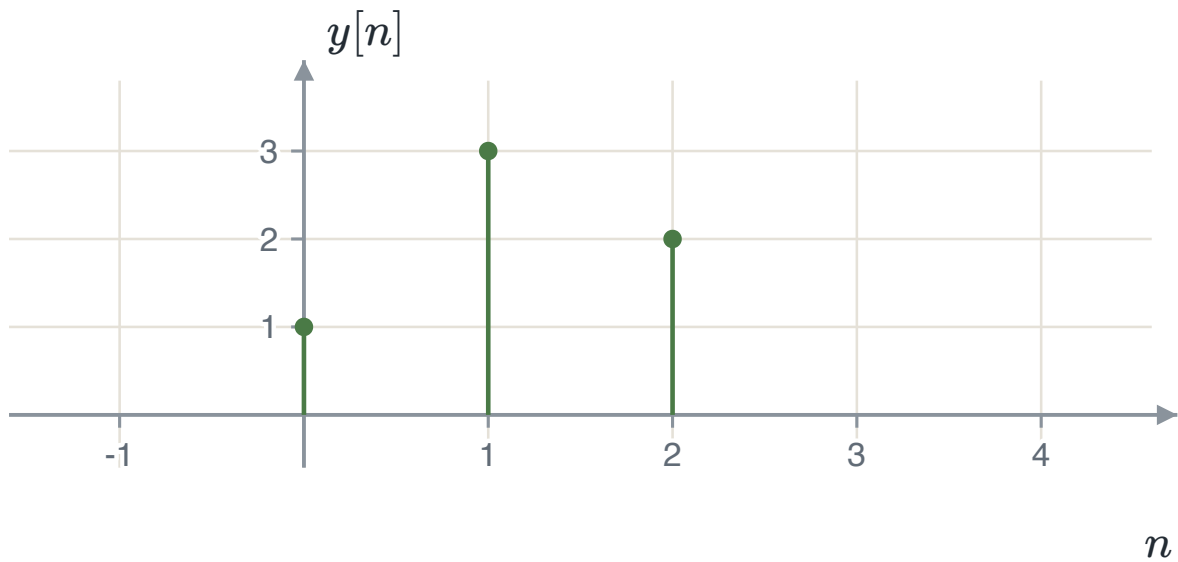
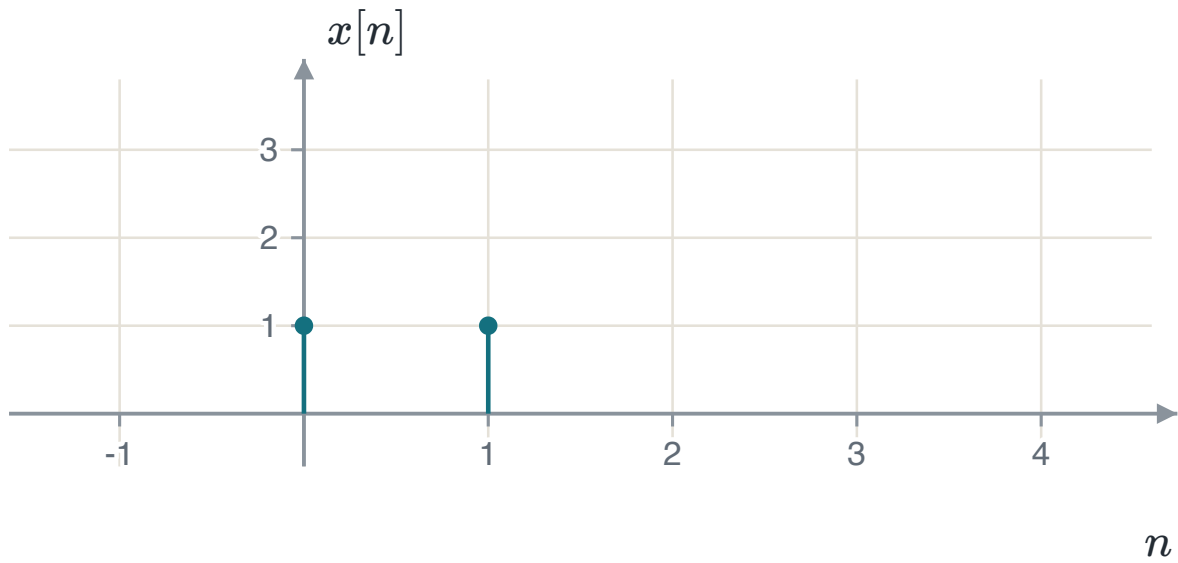
The $x(t)$ signal sketched below is an input to the LTI system with the impulse response $h(t)$, also sketched below. Compute and plot the output signal $y(t) = x(t) * h(t)$, using the continuous-time convolution.



- a) Write $h(t)$ in terms of step functions and give its running integral.
- b) Evaluate the convolution and plot $y(t)$.

D3-27

Consider a discrete-time *causal* LTI system where the output signal $y[n]$ is obtained when the input signal $x[n]$ is applied to this system. Both are plotted below.



- Calculate and plot the impulse response of this LTI system.
- Determine and plot the output signal if the input signal is defined as $x_2[n] = (-1)^n$ for $0 \leq n \leq 3$, and zero otherwise.

D3-28

Consider two discrete-time systems with input $x[n]$ and output $y[n]$, each defined by a difference equation and each initially at rest:

$$S_1 : y[n] = x[n] + 2y[n-1], \quad S_2 : y[n] = x[n] + \frac{1}{2}y[n-1].$$

- Plot the impulse response of S_1 .
- For the impulse response obtained in part (a), state whether the system is stable or not.
- Repeat both parts for S_2 , and say what decides the difference.

D3-29

An LTI system has the impulse response $h[n] = \delta[n] - \delta[n-2]$.

- Write the input-output difference equation of the system.
- For the input $x[n] = u[n] - u[n-4]$, compute and plot the output $y[n] = x[n] * h[n]$.
- State whether the system is causal and whether it is stable.

D3-30

Let $x(t) = u(t) - u(t-2)$ be an input to the LTI system which has the impulse response $h(t) = e^{-t}u(t)$.

- Identify the ranges of t over which the overlap changes, and write the convolution integral on each.
- Evaluate the integral and plot $y(t) = x(t) * h(t)$.
- State the value $y(t)$ approaches as $t \rightarrow \infty$ and explain it.

MODULE

4

Fourier Series

30 questions on fourier series. Write your reasoning in the space under each one.

D4-01

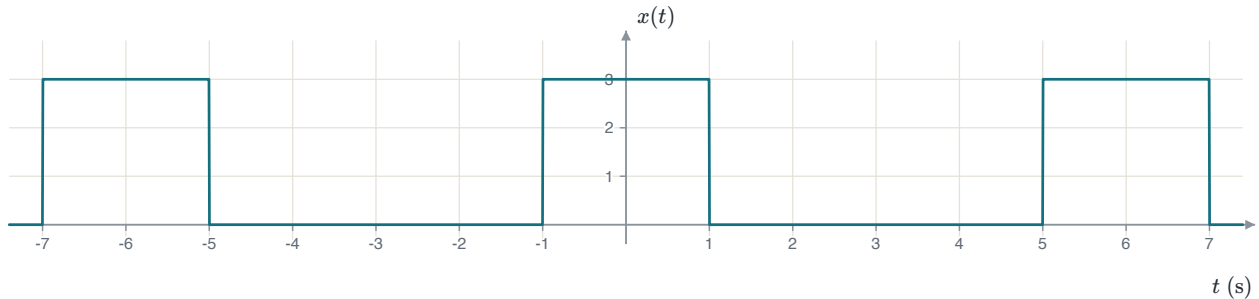
Let

$$x(t) = 4 + 2 \cos(2\pi t) + 6 \sin\left(\frac{2\pi}{3}t\right).$$

- Find the fundamental period T_0 and ω_0 , and state which harmonic each term is.
- Find a_k for every k , and plot the magnitude spectrum.
- Plot the phase spectrum.

D4-02

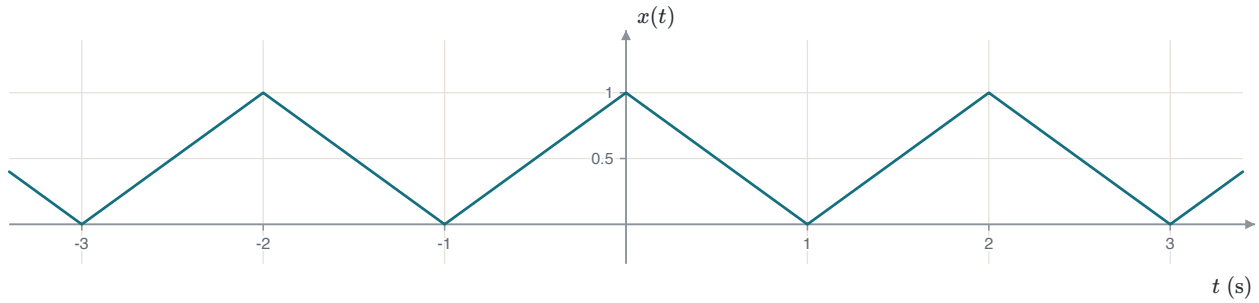
A periodic signal has $T_0 = 6$ and equals 3 for $|t| < 1$ and 0 for $1 < |t| < 3$ inside one period.



- Find a_0 by inspection.
- Find a_k for $k \neq 0$ using the analysis equation.
- Plot $|a_k|$ for $-6 \leq k \leq 6$ and say which harmonics are absent.

D4-03

A periodic signal has $T_0 = 2$ and equals $x(t) = 1 - |t|$ for $-1 \leq t \leq 1$ inside one period: a triangular pulse of height 1, with no jump anywhere, only a corner at $t = 0$ and at $t = \pm 1$.



- Find a_0 without integrating.
- Find a_k for $k \neq 0$ using integration by parts.
- Plot $|a_k|$ for $-7 \leq k \leq 7$ and state how fast the envelope decays.

D4-04

A real periodic signal $x(t)$ has fundamental angular frequency $\omega_0 = 2$ rad/s. Its only non-zero Fourier series coefficients are

$$a_0 = -1, \quad a_2 = 3j, \quad a_{-2} = -3j, \quad a_5 = 2e^{j\pi/4}, \quad a_{-5} = 2e^{-j\pi/4}.$$

- Using the conjugate-pair reassembly, write $x(t)$ as a sum of real cosines.
- Evaluate $x(0)$ from this real form.

D4-05

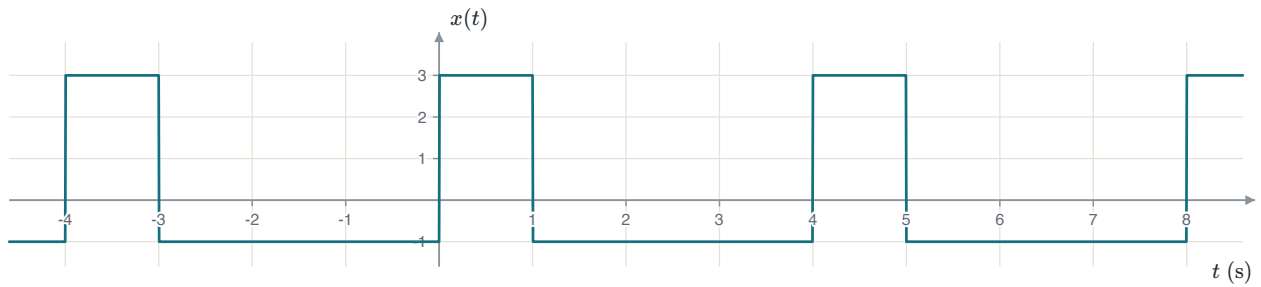
Let

$$x(t) = -3 + 4 \cos(5t) - 2 \sin(15t).$$

- Find the Fourier series coefficients a_k .
- Find the average power over one period using Parseval's relation.
- Confirm the same value using the direct amplitude rule for a sum of sinusoids.

D4-06

A periodic signal has $T_0 = 4$ and equals $x(t) = 3$ for $0 < t < 1$ and $x(t) = -1$ for $1 < t < 4$ inside one period.

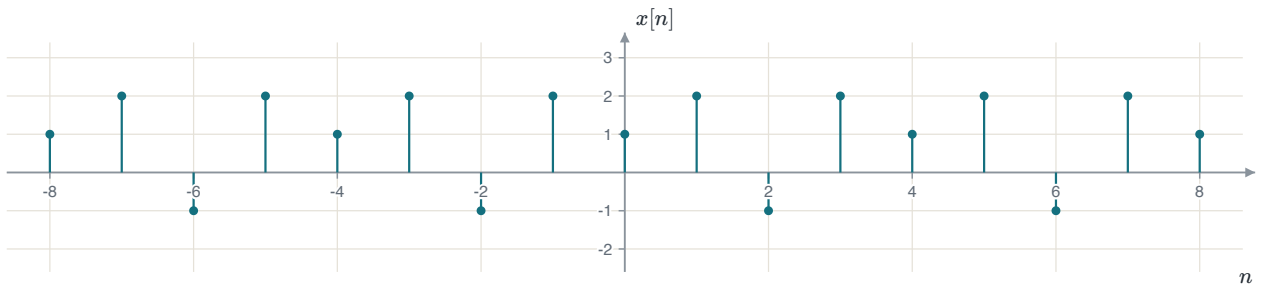


- Find the average power directly, using the period $0 \leq t < 4$.
- Recompute the average power using the period $-2 \leq t < 2$, and confirm the two agree.

D4-07

A periodic sequence has $N = 4$ and one period given by

$$x[0] = 1, \quad x[1] = 2, \quad x[2] = -1, \quad x[3] = 2.$$



- Find a_k for $k = 0, 1, 2, 3$ using the analysis sum.
- Find the average power directly in the time domain.
- Confirm the same value using Parseval's relation.

D4-08

A periodic signal has $T_0 = 2$ and equals $x(t) = 1 - |t|$ for $-1 \leq t \leq 1$ inside one period — the triangular pulse whose coefficients are $a_0 = \frac{1}{2}$ and $a_k = \frac{1 - (-1)^k}{(k\pi)^2}$ for $k \neq 0$.

- a) Find the average power P directly, by integrating x^2 over one period.
- b) Find P again as an infinite sum over the coefficients, using $\sum_{k \text{ odd}, k \geq 1} \frac{1}{k^4} = \frac{\pi^4}{96}$.
- c) Find the fraction of P already captured by the DC term and the first harmonic alone, and compare with a signal that has a jump.

D4-09

An LTI system has impulse response $h(t) = 5e^{-5t}u(t)$.

- a) Find the frequency response $H(j\omega)$.
- b) Find $|H(j\omega)|$ and $\angle H(j\omega)$ in closed form.
- c) Plot the magnitude spectrum, and evaluate it at $\omega = 5$ rad/s.

D4-10

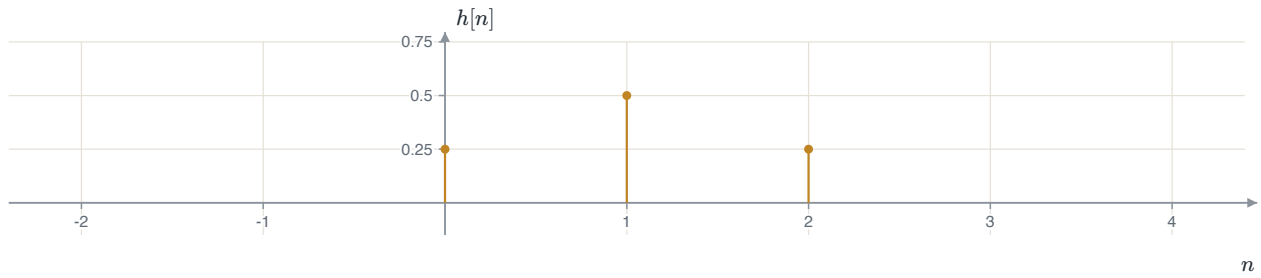
The periodic signal $x(t) = 2 + 3 \cos(4t)$ is the input to an LTI system with impulse response $h(t) = 3e^{-3t}u(t)$.

- a) Find the frequency response $H(j\omega)$.
- b) Find the Fourier series coefficients of $x(t)$.
- c) Find the Fourier series coefficients of the output $y(t)$.
- d) Write $y(t)$ in real form.

D4-11

A discrete-time LTI system has impulse response

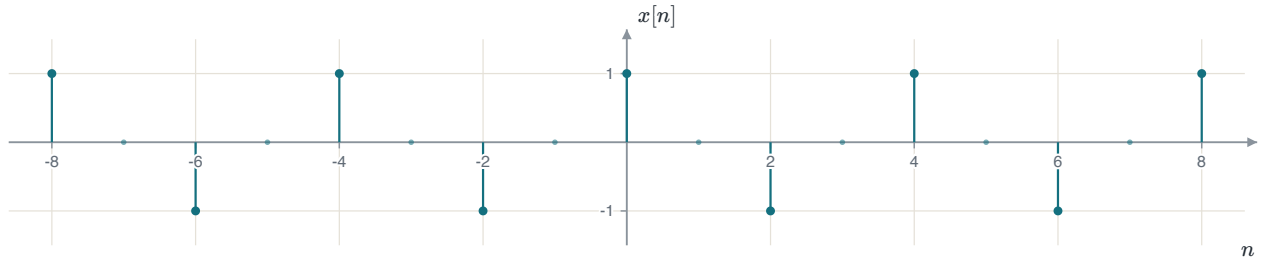
$$h[n] = 0.25 \delta[n] + 0.5 \delta[n - 1] + 0.25 \delta[n - 2].$$



- Find $H(e^{j\omega})$.
- Show that $|H(e^{j\omega})| = \cos^2(\omega/2)$, and plot the magnitude spectrum over one period.
- Evaluate H at $\omega = 0$ and at $\omega = \pi$, and say what kind of filter this is.

D4-12

The periodic sequence $x[n] = \cos(\frac{\pi}{2}n)$ is the input to a discrete-time LTI system with impulse response $h[n] = 0.6 \delta[n] + 0.4 \delta[n - 1]$.



- Find the fundamental period N and the coefficients a_k of $x[n]$.
- Find $H(e^{j\omega})$ and evaluate it at $\omega = \omega_0$.
- Find the output coefficients and write $y[n]$ in real form.
- Plot $y[n]$ as a stem for $-8 \leq n \leq 8$.

D4-13

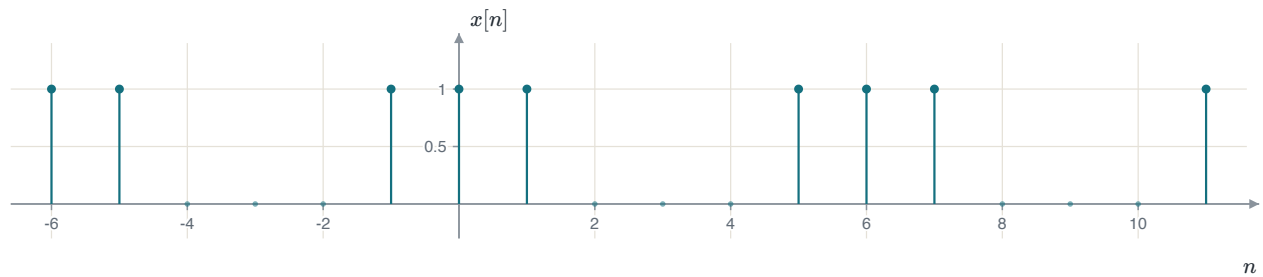
Let

$$x[n] = 3 \cos\left(\frac{\pi}{4}n\right) - \sin\left(\frac{2\pi}{3}n\right).$$

- Find the fundamental period N_0 and ω_0 , and state which harmonic each term is.
- Find every non-zero coefficient.
- Plot $|a_k|$ over one period, $-12 \leq k \leq 11$.

D4-14

A periodic sequence has $N = 6$ and equals 1, 1, 0, 0, 0, 1 for $n = 0, 1, 2, 3, 4, 5$ inside one period.



- List $x[n]$ for one full period and identify which samples are non-zero.
- Find a_k for $k = 0, 1, 2, 3$ using the analysis sum.
- Plot $|a_k|$ over one period and confirm a_0 equals the mean of the sequence.

D4-15

A real periodic sequence $x[n]$ has fundamental period $N = 5$. Its only non-zero coefficients are

$$a_0 = 2, \quad a_1 = 1 + j, \quad a_4 = 1 - j.$$

- Using the conjugate-pair reassembly, write $x[n]$ as a single real cosine plus a constant.
- Evaluate $x[0]$ directly from the three given coefficients, and from the real form.

D4-16

The period-6 sequence $x[0] = 1, x[1] = 1, x[2] = 0, x[3] = 0, x[4] = 0, x[5] = 1$ has Fourier series coefficients $a_0 = \frac{1}{2}, a_1 = a_5 = \frac{1}{3}, a_2 = a_4 = 0, a_3 = -\frac{1}{6}$.

- Reconstruct $x[0]$ using the synthesis sum over all six coefficients, and confirm it equals the given value.
- Reconstruct $x[2]$ the same way, using all six coefficients.
- Set a_3 to zero, keep the other five coefficients unchanged, and recompute $x[2]$. State the size of the resulting error.

D4-17

A real periodic signal $x(t)$ has fundamental angular frequency $\omega_0 = \pi$ rad/s and coefficients $a_1 = 2, a_2 = 1 - j$ (with $a_{-1} = 2, a_{-2} = 1 + j$ by conjugate symmetry). Let $y(t) = x(t - t_0)$ with $t_0 = 0.5$ s.

- State the general time-shift property, with the correct sign in the exponent.
- Compute b_1 and b_2 , and confirm $|b_k| = |a_k|$ for both.
- Compute $\angle b_2 - \angle a_2$ and confirm it equals $-2\omega_0 t_0$, modulo 2π .

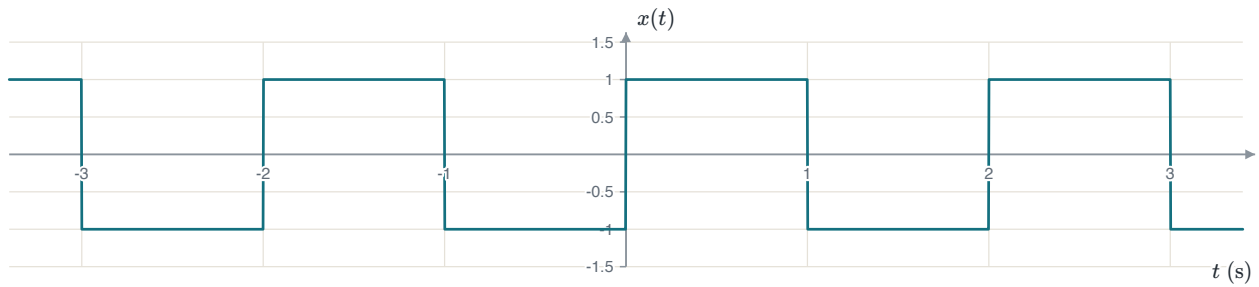
D4-18

A periodic signal $x(t)$ with fundamental angular frequency $\omega_0 = 5$ rad/s is real and even: $x(t) = x(-t)$ for every t , and $x(t)$ is real-valued.

- State what the time-reversal property alone gives for a_{-3} , in terms of a_3 .
- State what conjugation (realness) alone gives for a_{-3} , in terms of a_3 .
- Combine the two results to show that a_3 must be real, and decide whether $a_3 = 2 - 5j$ is a value a_3 could actually take.

D4-19

A periodic signal has $T_0 = 2$ and equals $x(t) = -1$ for $-1 < t < 0$ and $x(t) = 1$ for $0 < t < 1$ inside one period: a real, odd, bipolar square wave.



- State, without integrating, why $a_0 = 0$.
- Derive a_k for $k \neq 0$ using the analysis equation, splitting the integral at $t = 0$.
- Confirm every coefficient is purely imaginary, and plot the magnitude spectrum.

D4-20

A real periodic signal $x(t)$ has fundamental angular frequency $\omega_0 = 1$ rad/s. Its only non-zero coefficients are

$$a_0 = 1, \quad a_3 = 2e^{j\pi/3}, \quad a_5 = 1 - j.$$

- Use conjugate symmetry to state a_{-3} and a_{-5} .
- Write $x(t)$ in real closed form.
- Compute the average power two independent ways, and confirm they agree.

D4-21

Let

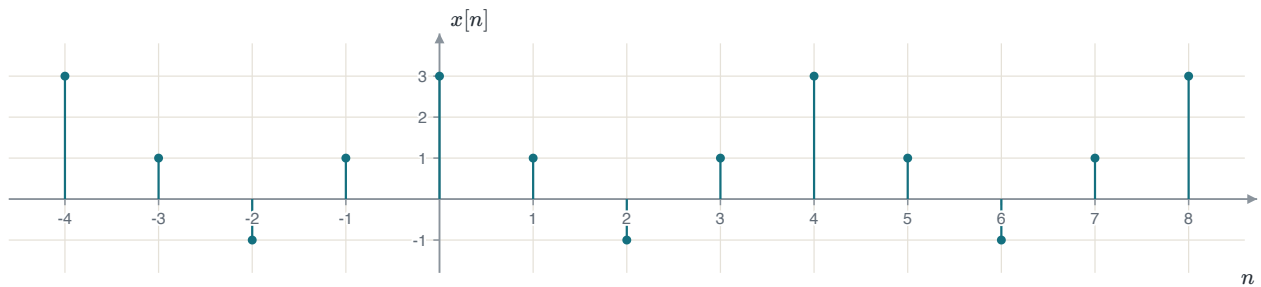
$$x[n] = 2 + (-1)^n \quad \text{and} \quad y[n] = \sin\left(\frac{4\pi}{5}n + \frac{\pi}{2}\right)$$

be two periodic signals.

- Determine the Fourier series coefficients and plot the magnitude of each coefficient for $x[n]$.
- Determine the Fourier series coefficients and plot the magnitude of each coefficient for $y[n]$.
- Let $z[n] = x[n]y[n]$ be a periodic signal with fundamental period 10. Determine the Fourier series coefficients and plot the magnitude of each coefficient for $z[n]$.

D4-22

Let $x[n]$ be a periodic signal with fundamental period $N = 4$, sketched below.



- Plot the Fourier series coefficients of $x[n]$.
- Compute the Fourier series coefficients and plot the magnitude of each coefficient for the signal $y[n] = x[n] - x[n-1]$.
- Calculate the average power of the $y[n]$ signal defined in part (b).

D4-23

Let $x(t)$ be a periodic signal defined as

$$x(t) = 6 \cos^2\left(\frac{\pi}{4}t\right) - 4 \sin\left(\frac{\pi}{3}t\right).$$

- a) Calculate the fundamental period of $x(t)$.
- b) Plot the magnitude and phase of each set of Fourier series coefficients of $x(t)$.
- c) Calculate the average power in one period of the periodic signal $x(t)$.

D4-24

Let $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 4k]$ be the input of a discrete-time LTI system with impulse response $h[n] = \left(\frac{1}{2}\right)^n u[n]$. Let $y[n] = x[n] * h[n]$ be the output signal.

- a) Determine the Fourier series coefficients and plot the magnitude of each coefficient for $x[n]$.
- b) Determine the frequency response of $h[n]$ and plot its magnitude spectrum.
- c) Determine the Fourier series coefficients and plot the magnitude of each coefficient for $y[n]$.

D4-25

Let

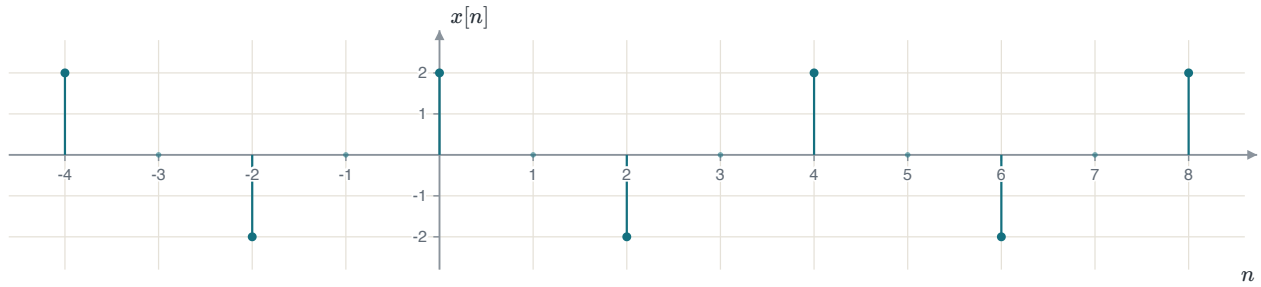
$$x(t) = 1 + 2 \cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right) + 3 \sin(\pi t)$$

be a periodic signal that is applied to a linear time-invariant (LTI) system with the impulse response $h(t) = \delta(t) - 2e^{-2t}u(t)$. At the output of the LTI system, the $y(t)$ signal is obtained.

- a) Determine the Fourier series coefficients and plot the magnitude of each coefficient for $x(t)$.
- b) Determine the frequency response of $h(t)$.
- c) Determine the Fourier series coefficients and plot the magnitude of each coefficient for $y(t)$.

D4-26

The periodic signal $x[n]$ sketched below is the input of a discrete-time LTI system with the impulse response $h[n] = u[n + 1] - u[n - 2]$. Let $y[n] = x[n] * h[n]$ be the output signal.



- Plot the Fourier series coefficients of $x[n]$.
- Determine the frequency response of $h[n]$.
- Plot the Fourier series coefficients of $y[n]$.

D4-27

Let

$$a_k = \begin{cases} 2, & k \text{ even} \\ 1, & k \text{ odd} \\ 0, & |k| \geq 3 \end{cases}$$

be the Fourier series coefficients of a continuous-time signal $x(t)$ which is periodic with 6 seconds.

- Find the Fourier series representation of $x(t)$, and write it as a sum of real cosines.
- Calculate the average power of $x(t)$.

D4-28

A periodic signal $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 6k]$ is the input of a discrete-time LTI system with the impulse response

$$h[n] = \frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}.$$

Let $y[n] = x[n] * h[n]$ be the output signal.

- Plot the Fourier series coefficients of $x[n]$.
- Plot the frequency response of $h[n]$.
- Calculate the average power in one period for the periodic signal $y[n]$.

D4-29

Let $x(t)$ be the periodic square wave of period $T_0 = 4$ seconds defined over one period by

$$x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & 1 < |t| < 2. \end{cases}$$

- Give the fundamental frequency ω_0 and the Fourier series coefficients a_k .
- State which coefficients vanish and say why.
- Calculate the average power of $x(t)$, and verify it against Parseval.

D4-30

Let

$$x(t) = 3 + 2 \cos(2t) + \cos(4t)$$

be the input to an LTI system with impulse response $h(t) = e^{-t}u(t)$, and let $y(t)$ be the output.

- Give the fundamental period of $x(t)$ and its Fourier series coefficients.
- Determine the frequency response $H(j\omega)$ and evaluate it at each harmonic present.
- Determine the Fourier series coefficients of $y(t)$ and plot their magnitudes.

MODULE

5

Continuous-Time Fourier Transform

30 questions on continuous-time fourier transform. Write your reasoning in the space under each one.

D5-01

Find the Fourier transform of

$$x(t) = e^{-4t}u(t).$$

- a) Compute $X(j\omega)$, stating the condition under which the integral converges.
- b) Evaluate $|X(j3)|$ and $\angle X(j3)$, and sketch $|X(j\omega)|$.
- c) Check $X(0)$ against the area of $x(t)$.

D5-02

Find the Fourier transform of

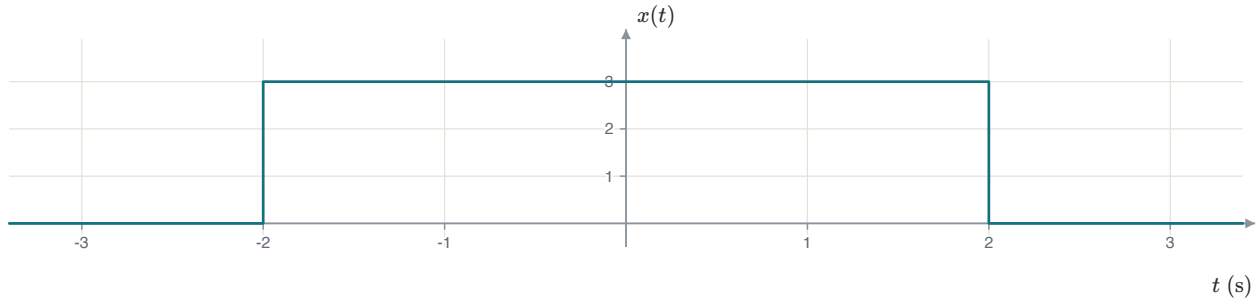
$$x(t) = t e^{-3t}u(t).$$

- a) Compute $X(j\omega)$ directly from the analysis equation.
- b) Give $|X(j\omega)|$, state where it peaks, and sketch it.
- c) Check $X(0)$ against the area of $x(t)$.

D5-03

A rectangular pulse of height 3 is defined by

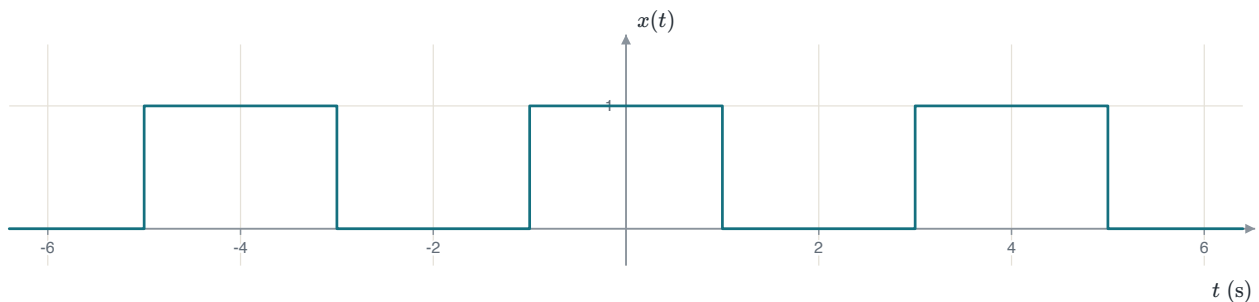
$$x(t) = \begin{cases} 3, & |t| < 2, \\ 0, & |t| > 2. \end{cases}$$



- Compute $X(j\omega)$ and write it using the unnormalised sinc.
- Give the locations of the zeros of $X(j\omega)$.
- Sketch $X(j\omega)$ and check $X(0)$ against the area of $x(t)$.

D5-04

A periodic rectangular pulse train has period $T_0 = 4$ and equals 1 on $|t| < 1$ inside each period, 0 elsewhere in the period.



- Find the Fourier series coefficients a_k of $x(t)$, using the analysis equation over one period.
- Give $X(j\omega)$ as a sum of impulses, and state the weight at $k = 0, \pm 1, \pm 2, \pm 3$.
- Sketch $X(j\omega)$ for $-2\pi \leq \omega \leq 2\pi$, showing the negative frequencies, and say which of the four impulses vanish.

D5-05

Find $x(t)$ given

$$X(j\omega) = \frac{1}{(j\omega + 2)(j\omega + 5)}.$$

- a) Split X into partial fractions.
- b) Give $x(t)$.
- c) Check your answer at $\omega = 0$.

D5-06

Find $x(t)$ given

$$X(j\omega) = \frac{j\omega + 6}{(j\omega + 1)(j\omega + 4)}.$$

- a) Split X into partial fractions.
- b) Give $x(t)$.
- c) Check your answer at $\omega = 0$, and state $x(0^+)$.

D5-07

Find $x(t)$ given

$$X(j\omega) = \frac{j\omega + 3}{(j\omega + 1)^2(j\omega + 2)}.$$

- a) Split X into partial fractions, using the repeated-pole rule at $j\omega = -1$.
- b) Give $x(t)$.
- c) Check your answer at $\omega = 0$ and at $t = 0^+$.

D5-08

A system is described by

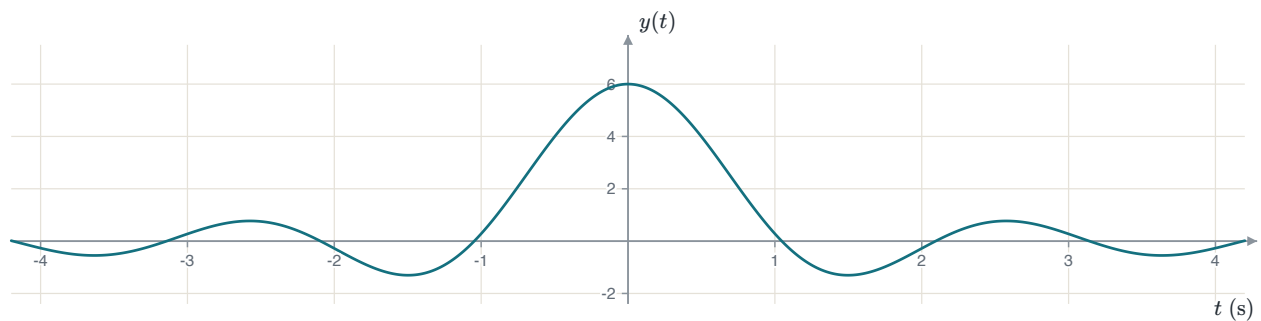
$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2x(t).$$

- Find the frequency response $H(j\omega) = Y(j\omega)/X(j\omega)$.
- Split H into partial fractions and give the impulse response $h(t)$.
- Check $H(0)$ against the area of $h(t)$.

D5-09

Let

$$y(t) = \frac{6 \sin(3t)}{3t} = \frac{2 \sin(3t)}{t}.$$



- Identify the known transform pair $y(t)$ is built from, using the unnormalised sinc, $\text{sinc}(\theta) = \sin \theta / \theta$.
- Apply duality to obtain $Y(j\omega)$, and sketch it.
- Check $Y(0)$ against the area of $y(t)$.

D5-10

Let

$$y(t) = e^{-3t}u(t), \quad z(t) = \frac{dy}{dt}.$$

- a) State the differentiation property and use it to find $Z(j\omega)$ from $Y(j\omega)$, without differentiating in time.
- b) Write $z(t)$ explicitly, using the product rule on $e^{-3t}u(t)$, and note the term that comes from the jump in y at $t = 0$.
- c) Check $Z(0)$ against the area of $z(t)$.

D5-11

Let $p(t) = 1$ for $|t| < 1$, 0 elsewhere, with $P(j\omega) = 2 \sin(\omega)/\omega$. Let

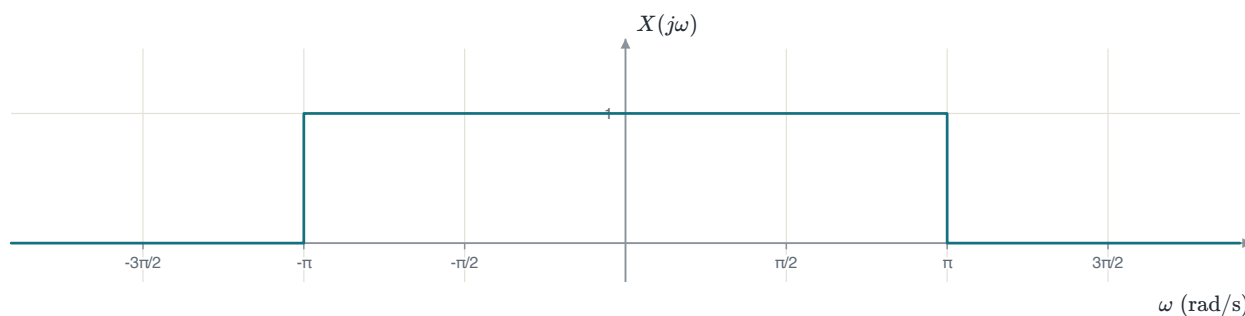
$$w(t) = p(3t - 6).$$

- a) Write $w(t)$ in the form $p(at - b)$ and identify a and b .
- b) Apply the scaling property, then the time-shift property, to find $W(j\omega)$ without integrating.
- c) Check $W(0)$ against the area of $w(t)$.

D5-12

Let $x(t) = \frac{\sin(\pi t)}{\pi t}$, whose transform is $X(j\omega) = 1$ on $|\omega| < \pi$, 0 elsewhere. Let

$$z(t) = e^{j3t}x(t).$$



- State the frequency-shift property in the form used here, naming what is fixed and what is being multiplied.
- Apply it to find $Z(j\omega)$ and sketch it, giving the band edges numerically.
- Check $z(0)$ against $x(0)$, and again against $\frac{1}{2\pi} \int Z(j\omega) d\omega$.

D5-13

Let $x(t) = 3e^{-2t}u(t)$.

- Compute the total energy directly in the time domain.
- Compute it again from $X(j\omega)$ using Parseval's relation.

D5-14

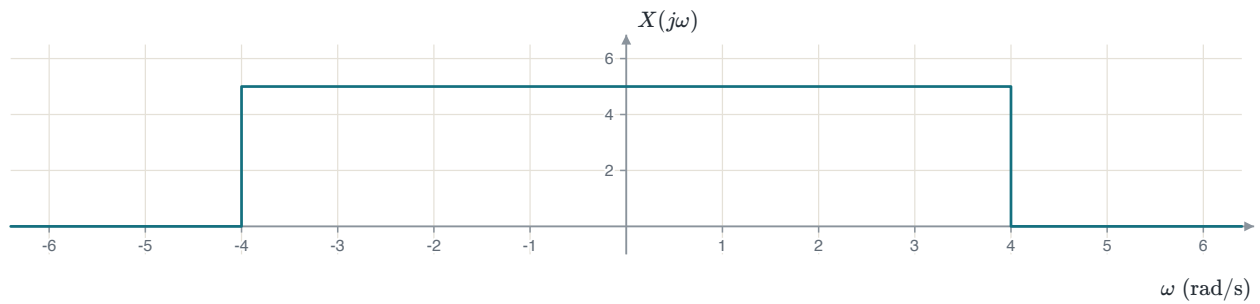
Let $y(t) = e^{-4|t|}$, whose transform is $Y(j\omega) = \frac{8}{16 + \omega^2}$.

- Compute the total energy of $y(t)$ in the time domain.

- Use Parseval's relation to evaluate $\int_{-\infty}^{\infty} \frac{d\omega}{(16 + \omega^2)^2}$.

D5-15

A signal has $X(j\omega) = 5$ for $|\omega| < 4$, 0 elsewhere.



- Compute the total energy of $x(t)$ using Parseval.
- Find $x(t)$ from the ideal low-pass pair, and check $x(0)$ against $\frac{1}{2\pi} \int X(j\omega) d\omega$.
- Find the fraction of the total energy that lies in $|\omega| < 2$.

D5-16

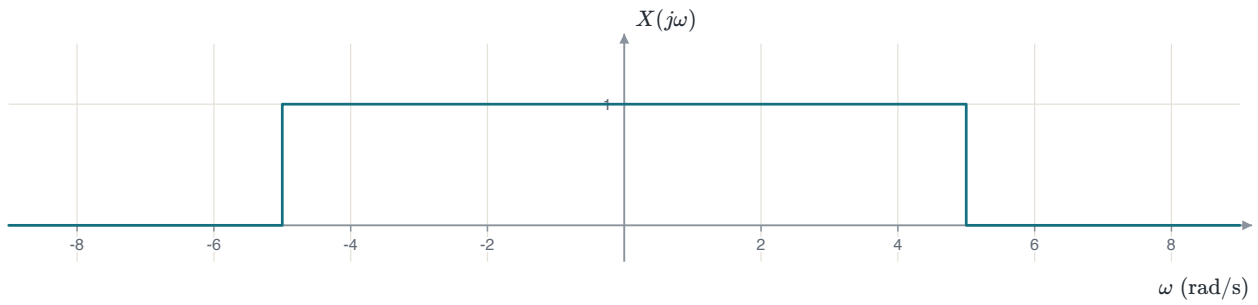
Let $x_1(t) = 2e^{-3t}u(t)$ and $x_2(t) = 2e^{3t}u(-t) = x_1(-t)$.

- Compute E_1 , the energy of $x_1(t)$, in the time domain.
- Find $X_2(j\omega)$ using the time-reversal property, and say what it implies about $|X_2(j\omega)|$ compared with $|X_1(j\omega)|$.
- State E_2 without a new frequency integral, then confirm it directly in the time domain.

D5-17

A signal has $X(j\omega) = 1$ for $|\omega| < 5$, 0 elsewhere. It is modulated by a carrier:

$$z(t) = x(t) \cos(20t).$$



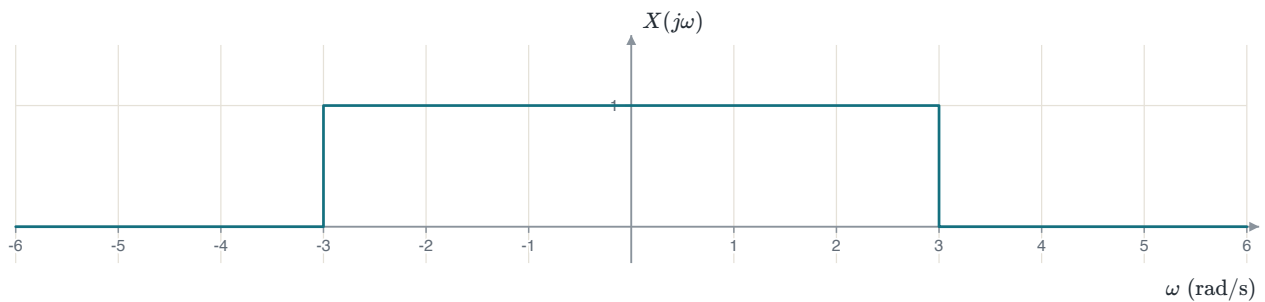
- Give $Z(j\omega)$.
- Sketch $Z(j\omega)$, showing the negative-frequency copy.
- State the general non-overlap condition on the carrier frequency, and verify it holds here.

D5-18

A baseband signal has $X(j\omega) = 1$ for $|\omega| < 3$, 0 elsewhere. It is sent through the chain

$$x(t) \xrightarrow{\times \cos(10t)} z(t) \xrightarrow{\times \cos(10t)} w(t) \xrightarrow{H(j\omega)} y(t),$$

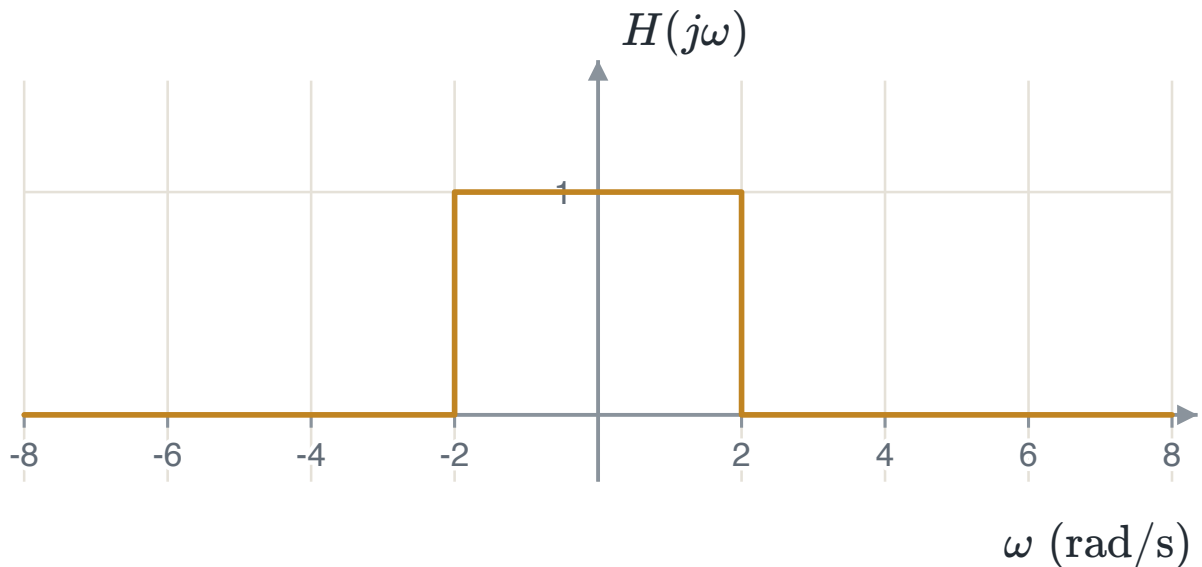
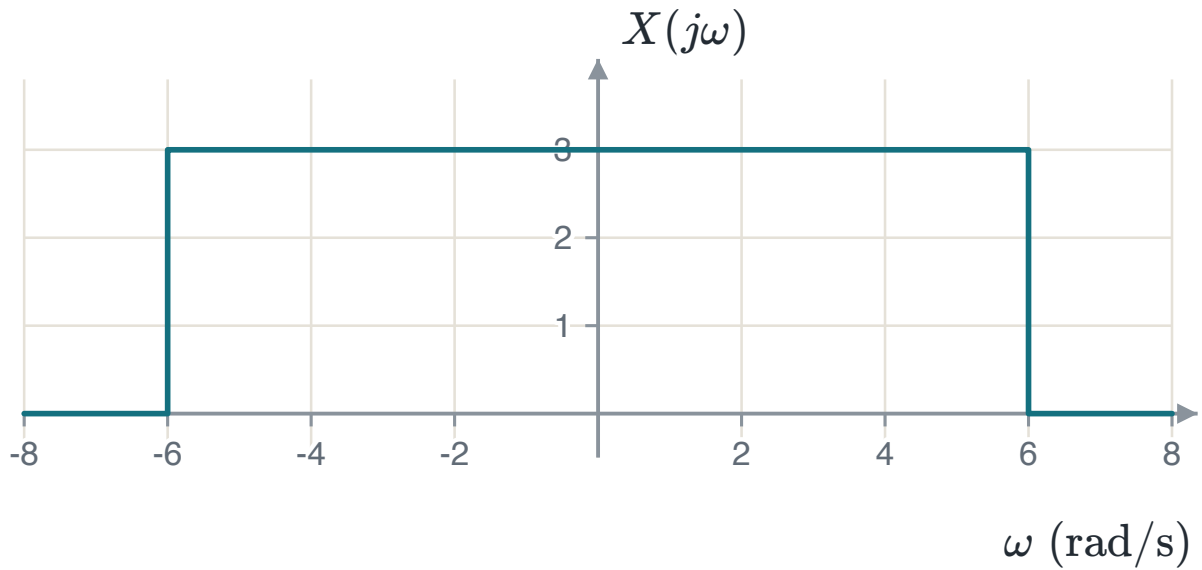
where the last stage is an ideal low-pass filter, $H(j\omega) = 2$ for $|\omega| < 5$, 0 elsewhere.



- Give $Z(j\omega)$, the spectrum after the first multiplication, and its two bands.
- Give $W(j\omega)$, the spectrum after the second multiplication, before the filter.
- Give $Y(j\omega)$ after the filter, and confirm it reproduces $X(j\omega)$ exactly.

D5-19

A signal has $X(j\omega) = 3$ for $|\omega| < 6$, 0 elsewhere. It is passed through an ideal low-pass filter, $H(j\omega) = 1$ for $|\omega| < 2$, 0 elsewhere.



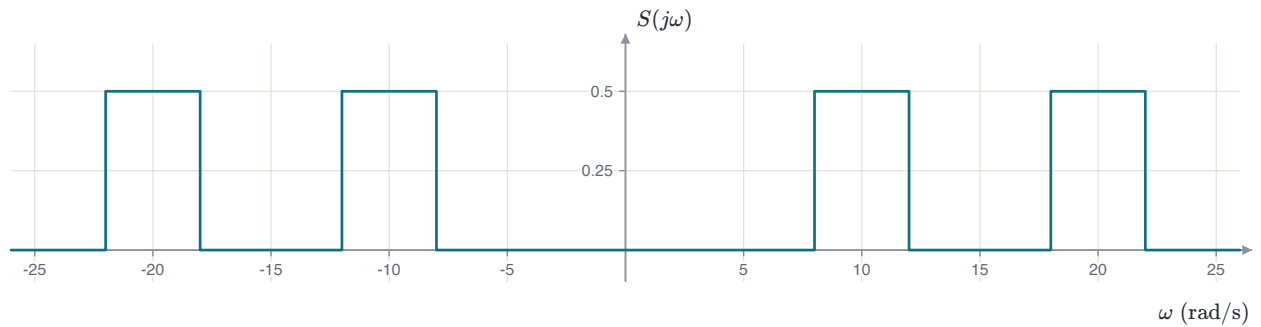
- Give $Y(j\omega) = X(j\omega)H(j\omega)$ and sketch it beside $X(j\omega)$.
- Find $x(t)$ and $y(t)$ in closed form.
- Compare $y(0)$ with $x(0)$, and relate the ratio to the two bandwidths.

D5-20

Two baseband signals, each with $X_1(j\omega) = X_2(j\omega) = 1$ for $|\omega| < 2$, are sent on one channel by two carriers:

$$s(t) = x_1(t) \cos(10t) + x_2(t) \cos(20t).$$

A receiver recovers $x_2(t)$ by multiplying $s(t)$ by $\cos(20t)$ and then applying an ideal low-pass filter, $H(j\omega) = 2$ for $|\omega| < 2$, 0 elsewhere.



- Give $S(j\omega)$, listing the centre and height of all four bands.
- After multiplying by $\cos(20t)$ a second time, give the part of the resulting spectrum lying in $|\omega| < 2$, and say which baseband signal it came from.
- Apply the filter and confirm the output equals $X_2(j\omega)$ exactly. State the nearest surviving band and how far it sits from the filter's cut-off.

D5-21

Determine the Fourier transforms of the following signals and plot their corresponding magnitude spectra.

- $x(t) = \cos(2t) u(t)$.
- $x(t) = u(t + 3) - u(t - 3)$.
- $x(t) = e^{-t(2+j10\pi)} u(t)$.

D5-22

Please solve the following problems. *Hint: if $x(t) \xrightarrow{\mathcal{F}} X(j\omega) = X(\omega)$, then $X(t) \xrightarrow{\mathcal{F}} 2\pi x(-\omega)$.*

- a) Determine the Fourier transform of $x(t) = e^{-3|t|}$.
- b) Determine the Fourier transform of $y(t) = \frac{6}{9 + t^2}$.
- c) Calculate the total energy in $y(t)$, which is defined in part (b).

D5-23

Please solve the following problems.

- a) Determine the inverse Fourier transform of $X(j\omega) = \frac{j\omega + 7}{(j\omega)^2 + 5j\omega + 6}$.
- b) Determine the Fourier transform of $x(t) = e^{-3(t+2)} \cos(4t) u(t + 2)$.
- c) Determine the Fourier transform of $x(t) = \sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k \delta(t - 2k)$.

D5-24

Let

$$x(t) = \cos(t) + \cos(3t) + 2 \cos(5t)$$

be an input to the LTI system which has the following impulse response,

$$h(t) = \pi \frac{\sin(2t)}{\pi t} \frac{\sin(4t)}{\pi t}.$$

Let $y(t) = x(t) * h(t)$ be the output signal.

- a) Plot the Fourier transform of $x(t)$.
- b) Plot the Fourier transform of $h(t)$.
- c) Plot the Fourier transform of $y(t)$, and give $y(t)$.

D5-25

Let

$$x(t) = \frac{1 - e^{-j8\pi t}}{j2\pi t}$$

be an input to the following communication chain, where $\times$ shows the multiplication operator and the LTI system has the impulse response $h(t) = e^{-j3\pi t} \frac{\sin(2\pi t)}{\pi t}$. The input is multiplied by $\cos(6\pi t)$ to give $y(t)$, and $y(t)$ is filtered by $h(t)$ to give $z(t)$.

- a) Plot the Fourier transform of $x(t)$. *Hint: can you write $x(t)$ as $e^{jW_1 t} \frac{\sin(W_2 t)}{\pi t}$ for some W_1, W_2 ?*
- b) Plot the Fourier transform of $y(t)$.
- c) Plot the Fourier transform of $z(t)$.

D5-26

Consider the signal

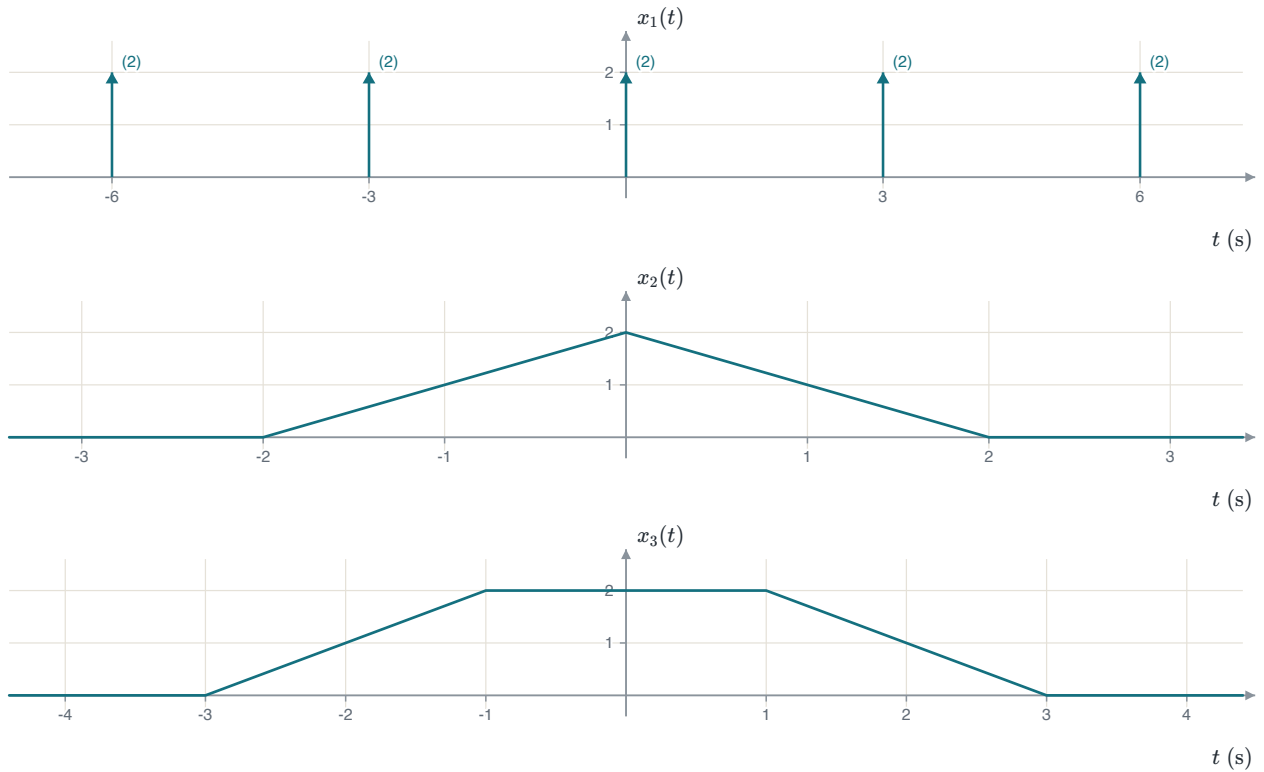
$$x(t) = \left(\frac{\sin(3t)}{\pi t} \right)^2 (1 + e^{j12t}),$$

which is input to an LTI system with the impulse response $h(t) = \frac{\sin(5t)}{\pi t} e^{j5t}$. At the output of the LTI system, the $y(t)$ signal is observed.

- a) Plot the Fourier transform of $x(t)$.
- b) Plot the Fourier transform of $h(t)$.
- c) Plot the Fourier transform of $y(t)$.

D5-27

Consider the three signals plotted below, where $x_1(t)$ is periodic and its plot shows the areas under its impulses, while $x_2(t)$ and $x_3(t)$ are aperiodic.



- Determine the Fourier transform of $x_1(t)$.
- Determine the Fourier transform of $x_2(t)$.
- Determine the Fourier transform of $x_3(t)$. *Hint: consider building it from rectangles.*

D5-28

Let $x(t) = 1 + 3 \cos(2\pi t)$ be a periodic signal, and let $s(t) = x(t)c(t)$ where $c(t) = \cos(10\pi t)$.

- Determine the Fourier transform of $x(t)$.
- Plot the Fourier transform of $s(t)$.
- State the smallest carrier frequency for which the shifted copies would not overlap, and say what happens below it.

D5-29

Determine the Fourier transforms of the following signals.

- a) $x(t) = e^{-3t}u(t-2)$.
- b) $x(t) = te^{-3t}u(t)$.
- c) $x(t) = [u(t+2) - u(t-2)] * [u(t+2) - u(t-2)]$, where $*$ is the convolution operator.

D5-30

Let $x(t) = \frac{\sin(4t)}{\pi t}$, and let $y(t) = x(t) \cos(6t)$.

- a) Plot the Fourier transform of $x(t)$ and calculate its total energy.
- b) Plot the Fourier transform of $y(t)$.
- c) Calculate the total energy of $y(t)$ and compare it with that of $x(t)$.

MODULE

6

Discrete-Time Fourier Transform

30 questions on discrete-time fourier transform. Write your reasoning in the space under each one.

D6-01

Find the discrete-time Fourier transform of

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$

- a) Compute $X(e^{j\omega})$ in closed form, and state the condition it needs to converge.
- b) Evaluate $|X(e^{j\omega})|$ at $\omega = 0$ and $\omega = \pi$.
- c) Plot $|X(e^{j\omega})|$ over one period.

D6-02

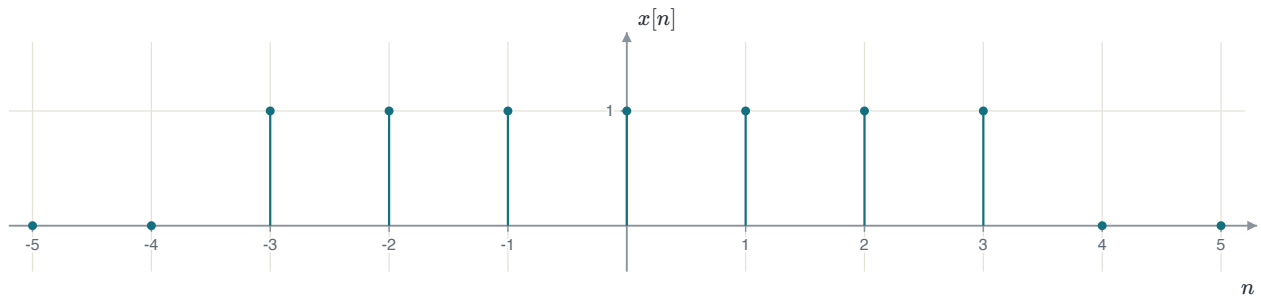
Find the discrete-time Fourier transform of

$$x[n] = 3 \left(\frac{1}{4}\right)^{|n|}.$$

- a) Split the sum at $n = 0$ and derive $X(e^{j\omega})$ in closed form.
- b) Evaluate $X(e^{j0})$ and $X(e^{j\pi})$.
- c) State whether $X(e^{j\omega})$ is real for every ω , and why.

D6-03

The sequence $x[n]$ shown below equals 1 for $-3 \leq n \leq 3$ and is zero elsewhere.



- State the support of $x[n]$ and give $X(e^{j\omega})$ as a ratio of sines.
- Give $X(e^{j0})$ and locate the zeros of X inside $-\pi \leq \omega \leq \pi$.
- Plot $|X(e^{j\omega})|$ over one period, marking the zeros.

D6-04

Find the discrete-time Fourier transform of

$$x[n] = 3 \left(\frac{1}{3}\right)^{n-1} u[n-1].$$

- Compute $X(e^{j\omega})$ directly from the analysis sum.
- Give $X(e^{j\omega})$ a second way, using the time-shift property on $w[n] = 3 \left(\frac{1}{3}\right)^n u[n]$, and confirm the two routes agree.
- Evaluate $X(e^{j0})$ and check it against the sum of the sequence.

D6-05

Plot the discrete-time Fourier transform of

$$x[n] = 4 \cos\left(\frac{\pi}{3}n\right)$$

on $-\pi \leq \omega \leq \pi$.

- Give $X(e^{j\omega})$ as a sum of impulses on one period.
- Mark the impulse weights and locations on a plot.
- Confirm the weights by synthesising $x[0]$ from the two impulses.

D6-06

Plot the discrete-time Fourier transform of

$$x[n] = 2 + 3 \cos\left(\frac{2\pi}{5}n\right)$$

on $-\pi \leq \omega \leq \pi$.

- a) Give $X(e^{j\omega})$ as a sum of impulses on one period.
- b) State the weight of the impulse at $\omega = 0$, and say why it differs from the weight the cosine term contributes at either of its two frequencies.
- c) Plot the spectrum, marking every weight.

D6-07

Find the discrete-time Fourier transform of

$$x[n] = 5 \sin\left(\frac{3\pi}{4}n\right).$$

- a) Derive the impulse pair for $\sin(\omega_0 n)$ from the impulse pair for $e^{j\omega_0 n}$, using Euler's relation.
- b) Give $X(e^{j\omega})$ for this $x[n]$.
- c) Check the result by synthesising $x[1]$.

D6-08

Plot the discrete-time Fourier transform of

$$x[n] = 2 \cos\left(\frac{3\pi}{2}n\right) + \cos\left(\frac{9\pi}{4}n\right)$$

on $-\pi \leq \omega \leq \pi$.

- a) Reduce each frequency into $-\pi \leq \omega \leq \pi$, and confirm the reduction leaves $x[n]$ unchanged at $n = 2$.
- b) Give $X(e^{j\omega})$ as a sum of impulses over one period.
- c) Plot the spectrum, and check the total weight against $x[0]$.

D6-09

A discrete-time LTI system has impulse response $h[n] = \left(\frac{1}{2}\right)^n u[n]$ and input $x[n] = \cos\left(\frac{\pi}{2}n\right)$.

- Give $H(e^{j\omega})$.
- Evaluate H at $\omega = \frac{\pi}{2}$, and give $y[n]$ in amplitude-and-phase form.
- Check $|H(e^{j\pi/2})|$ using $|H(e^{j\omega})|^2 = H(e^{j\omega})H^*(e^{j\omega})$.

D6-10

A discrete-time LTI system has impulse response

$$h[n] = \delta[n] - \delta[n - 1].$$

- Give $H(e^{j\omega})$, and show that $|H(e^{j\omega})| = 2|\sin(\omega/2)|$.
- Give $|H|$ at $\omega = 0$ and $\omega = \pi$, and say what kind of filter this is.
- Plot $|H(e^{j\omega})|$ over one period.

D6-11

A discrete-time LTI system has impulse response $h[n] = \left(\frac{1}{4}\right)^n u[n]$ and input $x[n] = \left(\frac{1}{2}\right)^n u[n]$.

- Give $X(e^{j\omega})$, $H(e^{j\omega})$ and $Y(e^{j\omega})$.
- Recover $y[n]$ by partial fractions.
- Check $y[0]$ and $y[1]$ directly against the convolution sum.

D6-12

A two-tap averaging filter has impulse response

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1],$$

and its input is $x[n] = \cos\left(\frac{\pi}{3}n\right)$.

- a) Write $H(e^{j\omega}) = e^{-j\omega/2} \cos(\omega/2)$, and evaluate it at $\omega = \frac{\pi}{3}$.
- b) Give $y[n]$ in amplitude-and-phase form.
- c) Give $Y(e^{j\omega})$ as a pair of weighted impulses, and plot $|Y(e^{j\omega})|$.

D6-13

Find $x[n]$ when

$$X(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{3}, \\ 0, & \frac{\pi}{3} < |\omega| \leq \pi, \end{cases}$$

repeated with period 2π .

- a) Compute $x[n]$ from the synthesis integral, for $n \neq 0$.
- b) Give $x[0]$ two ways: as the limit of your part (a) formula, and as the mean of the spectrum over one period.
- c) Plot $x[n]$ for $-8 \leq n \leq 8$.

D6-14

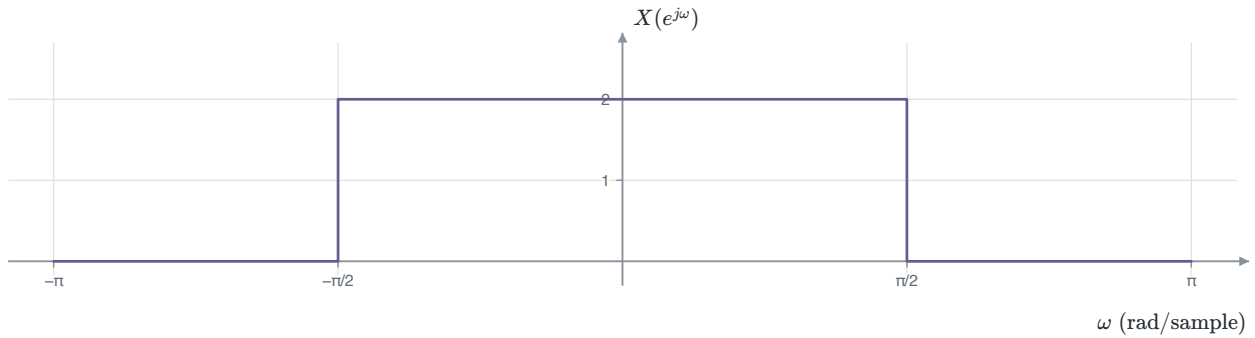
Find $x[n]$ when

$$X(e^{j\omega}) = 4 - e^{-j\omega} + 2e^{-j3\omega}.$$

- a) Give $x[n]$.
- b) State the support of $x[n]$ and give $\sum_n x[n]$.

D6-15

The spectrum $X(e^{j\omega})$ shown below is one period of a periodic function.



- Read the height and the cut-off frequency of $X(e^{j\omega})$ from the figure, and write it as a formula.
- Compute $x[n]$ for $n \neq 0$ from the synthesis integral.
- Give $x[0]$, and check it against the mean of the spectrum read directly from the figure.

D6-16

Find $x[n]$ when

$$X(e^{j\omega}) = e^{-j\omega} (1 + \cos \omega).$$

- Expand $X(e^{j\omega})$ into a sum of terms $e^{-j\omega n_0}$.
- Read off $x[n]$ from the expansion.
- Confirm the result at $\omega = 0$ and at $\omega = \pi$, against the original form.

D6-17

Let $x[n]$ be a real sequence with transform $X(e^{j\omega}) = \sum_n x[n] e^{-j\omega n}$.

- Prove that $X(e^{-j\omega}) = X^*(e^{j\omega})$, directly from the analysis sum.
- For $x[n] = \left(\frac{1}{3}\right)^n u[n]$, give $X(e^{j\pi/3})$ in the form $p + jq$, and give $X(e^{-j\pi/3})$ the same way by direct substitution.
- Confirm that the two values of part (b) are complex conjugates, and say why one half of a period is enough to describe the spectrum of a real sequence but not of a complex one.

D6-18

Let $x[n] = \delta[n+1] - \delta[n-1]$.

- a) Give $X(e^{j\omega})$, and confirm $X(e^{j0}) = 0$.
- b) Apply the differencing property to find $W(e^{j\omega})$ for $w[n] = x[n] - x[n-1]$, and confirm it against $w[n]$ written out directly.
- c) Apply the accumulation property to find $Y(e^{j\omega})$ for $y[n] = \sum_{k=-\infty}^n x[k]$, and confirm it against $y[n]$ written out directly.

D6-19

Let $x[n] = 2 \left(\frac{1}{2}\right)^n u[n]$.

- a) Compute the energy of $x[n]$ directly, in the time domain.
- b) Write Parseval's relation for $X(e^{j\omega})$, and state what $|X(e^{j\omega})|^2$ represents.
- c) Using $\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d\omega}{b - \cos \omega} = \frac{1}{\sqrt{b^2 - 1}}$ for $b > 1$, evaluate the frequency-side integral and confirm it matches part (a).

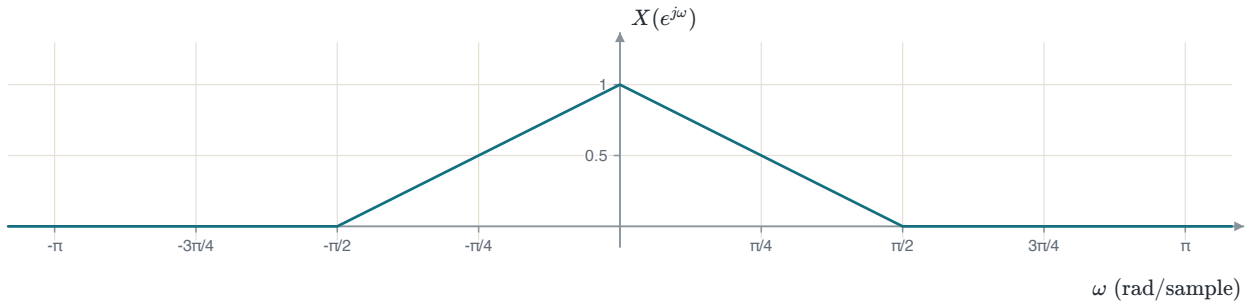
D6-20

Let $x[n]$ be an arbitrary sequence with transform $X(e^{j\omega}) = \sum_n x[n]e^{-j\omega n}$, convergent for every ω .

- a) Prove that $X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$, directly from the analysis sum.
- b) Two signal generators produce the sequences $\cos(2.3n)$ and $\cos((2.3 - 2\pi)n)$. Determine whether the two generators produce the same sequence, and say what part (a) implies about telling them apart from their output alone.
- c) State the highest discrete-time frequency that exists, and explain in one sentence why continuous time has no such bound.

D6-21

Determine the discrete-time Fourier transforms of the following signals. In parts (b) and (c), $\mathcal{F}\{x[n]\}$ is the triangular spectrum plotted below: it equals 1 at $\omega = 0$ and falls linearly to zero at $\omega = \pm\frac{\pi}{2}$.



- a) $z[n] = u[-n + 4] - u[-n - 1]$.
 b) $y[n] = x[n]p[n]$, where $p[n] = \sum_{k=-\infty}^{\infty} \delta[n - 4k]$.
 c) $y[n] = x[n]p[n]$, where $p[n] = (-1)^n$.

D6-22

Please solve the following problems, giving every answer in $-\pi \leq \omega \leq \pi$.

- a) Determine the DTFT of $x[n] = \left(\frac{1}{3}\right)^{|n|}$.
 b) Plot the DTFT of $x[n] = \cos\left(\frac{5\pi}{2}n\right) + 2\cos\left(\frac{4\pi}{3}n\right)$.
 c) Plot the DTFT of $x[n] = \frac{\cos(\pi n)}{\pi n} \sin\left(\frac{\pi}{3}n\right)$.

D6-23

Consider the discrete-time signal

$$x[n] = \frac{\sin\left(\frac{3\pi}{5}n\right) + 2\sin\left(\frac{\pi}{5}n\right)}{\pi n},$$

which is input to an LTI system with the impulse response

$$h[n] = 2 \frac{\cos(\pi n)}{\pi n} \sin\left(\frac{4\pi}{5}n\right).$$

At the output of the LTI system, the $y[n]$ signal is observed.

- a) Plot the DTFT of $x[n]$ in $-\pi \leq \omega \leq \pi$.
- b) Plot the DTFT of $h[n]$ in $-\pi \leq \omega \leq \pi$.
- c) Plot the DTFT of $y[n]$ in $-\pi \leq \omega \leq \pi$.

D6-24

Determine the discrete-time Fourier transforms of the following signals.

- a) $x[n] = \left(\frac{1}{2}\right)^n u[n-1]$.
- b) $x[n] = u[n+2] - u[n-3]$.
- c) $x[n] = \left(\frac{1}{2}\right)^{|n|}$.

D6-25

A sequence $x[n]$ has the discrete-time Fourier transform $X(e^{j\omega})$. Express the DTFT of each of the following in terms of $X(e^{j\omega})$.

- a) $y[n] = x[n-3]$.
- b) $y[n] = (-1)^n x[n]$.
- c) $y[n] = x[n] - x[n-1]$, and state the value of Y at $\omega = 0$.

D6-26

Determine the inverse discrete-time Fourier transform of each of the following spectra, each specified over $-\pi \leq \omega \leq \pi$ and repeating with period 2π .

a) $X(e^{j\omega}) = 1$ for $|\omega| < \frac{\pi}{4}$ and zero otherwise.

b) $X(e^{j\omega}) = \cos(2\omega)$.

c) $X(e^{j\omega}) = \frac{1}{(1 - \frac{1}{3}e^{-j\omega})^2}$.

D6-27

Let $x[n] = \frac{\sin(\frac{\pi}{3}n)}{\pi n}$.

a) Give $X(e^{j\omega})$ over $-\pi \leq \omega \leq \pi$ and state $x[0]$.

b) Calculate the total energy of $x[n]$ using Parseval.

c) Let $y[n] = (-1)^n x[n]$. Give $Y(e^{j\omega})$ and its energy.

D6-28

An LTI system has the impulse response $h[n] = (\frac{1}{3})^n u[n]$ and is driven by the input $x[n] = (\frac{1}{2})^n u[n]$.

a) Give $X(e^{j\omega})$ and $H(e^{j\omega})$.

b) Give $Y(e^{j\omega})$ and find $y[n]$ by partial fractions.

c) Verify $y[0]$ and $y[1]$ against the convolution sum.

D6-29

Two filters are defined by

$$h_1[n] = \frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}, \quad h_2[n] = \delta[n] - \frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}.$$

- a) Plot $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ over $-\pi \leq \omega \leq \pi$.
- b) Determine the impulse response of the parallel connection $h_1[n] + h_2[n]$ and its frequency response.
- c) Determine the frequency response of the series connection $h_1[n] * h_2[n]$, and say what the connection does.

D6-30

Let $p[n] = \sum_{k=-\infty}^{\infty} \delta[n - 5k]$ be a periodic impulse train, and let $x[n]$ be a sequence whose DTFT is 1 for $|\omega| < \frac{\pi}{5}$ and zero elsewhere in $-\pi \leq \omega \leq \pi$.

- a) Determine the DTFT of $p[n]$.
- b) Determine and plot the DTFT of $y[n] = x[n]p[n]$.
- c) State whether $x[n]$ can be recovered from $y[n]$ by an ideal low-pass filter, and give the filter if it can.

MODULE

7

Sampling and Aliasing

30 questions on sampling and aliasing. Write your reasoning in the space under each one.

D7-01

Find the Nyquist rate for

$$x(t) = \cos(40\pi t) + \cos(90\pi t).$$

- a) List the frequencies present and give ω_M .
- b) Give the Nyquist rate in rad/s and in hertz, and the largest usable sampling period.

D7-02

Find the Nyquist rate for

$$x(t) = \cos(30\pi t) \cos(80\pi t).$$

- a) Rewrite $x(t)$ as a sum of cosines.
- b) Give ω_M and the Nyquist rate, in rad/s and in hertz.

D7-03

Find the Nyquist rate for

$$x(t) = \sin(50\pi t) \cos(30\pi t).$$

- a) Rewrite $x(t)$ as a sum of sines, using $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$.
- b) Give ω_M and the Nyquist rate, in rad/s and in hertz.

D7-04

Let

$$x(t) = \cos(20\pi t) + \cos(70\pi t), \quad y(t) = x(t) + x^2(t).$$

- a) List every frequency present in $y(t)$.
 b) Give ω_M and the Nyquist rate for $y(t)$, in rad/s and in hertz.

D7-05

Find the Nyquist rate for

$$x(t) = \frac{\sin(300\pi t)}{\pi t}.$$

- a) Give $X(j\omega)$.
 b) Give ω_M and the Nyquist rate, in rad/s and in hertz.

D7-06

Find the Nyquist rate for

$$x(t) = \left[\frac{\sin(100\pi t)}{\pi t} \right]^2.$$

- a) Describe $X(j\omega)$: its shape, its peak, and its edges.
 b) Give ω_M and the Nyquist rate, in rad/s and in hertz.
 c) Sketch $X(j\omega)$.

D7-07

Find the Nyquist rate for

$$x(t) = \frac{\sin(180\pi t)}{\pi t} + \frac{\sin(500\pi t)}{\pi t}.$$

- a) Describe $X(j\omega)$ and sketch it.
- b) Give ω_M and the Nyquist rate, and say which term determines it.

D7-08

Find the Nyquist rate for

$$x(t) = \left[\frac{\sin(140\pi t)}{\pi t} \right]^2 + \frac{\sin(90\pi t)}{\pi t}.$$

- a) Give the bandwidth of each of the two terms separately.
- b) Give ω_M and the Nyquist rate for the sum, in rad/s and in hertz.

D7-09

Two signals are band-limited: $X_1(j\omega) = 0$ for $|\omega| > 70\pi$ and $X_2(j\omega) = 0$ for $|\omega| > 190\pi$. Define

$$y(t) = x_1(t) * x_2(t).$$

- a) Give the band-limit of $y(t)$.
- b) Give the Nyquist rate for $y(t)$, and justify the answer in one sentence.

D7-10

With the same two signals — $X_1(j\omega) = 0$ for $|\omega| > 70\pi$ and $X_2(j\omega) = 0$ for $|\omega| > 190\pi$ — define

$$z(t) = x_1(t) x_2(t).$$

- a) Give the band-limit of $z(t)$.
- b) Give the Nyquist rate for $z(t)$, and contrast the answer with the convolution case.

D7-11

Let

$$z(t) = \frac{\sin(60\pi t)}{\pi t} \cos(400\pi t).$$

- a) Describe $Z(j\omega)$, giving the edges of every band it occupies.
- b) Give ω_M and the Nyquist rate, in rad/s and in hertz.

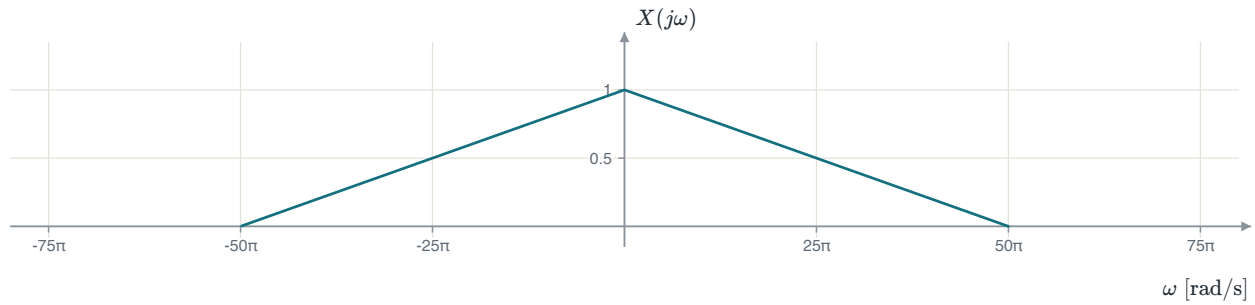
D7-12

Two signals are band-limited: $X_1(j\omega) = 0$ for $|\omega| > 50\pi$ and $X_2(j\omega) = 0$ for $|\omega| > 130\pi$. Their product $w(t) = x_1(t)x_2(t)$ is to be sampled and reconstructed with a filter whose transition band is $\omega_g = 20\pi$ rad/s wide.

- a) Give ω_M for $w(t)$.
- b) Give the minimum sampling rate when an ideal reconstruction filter is assumed.
- c) Give the minimum sampling rate that leaves room for the stated guard band.

D7-13

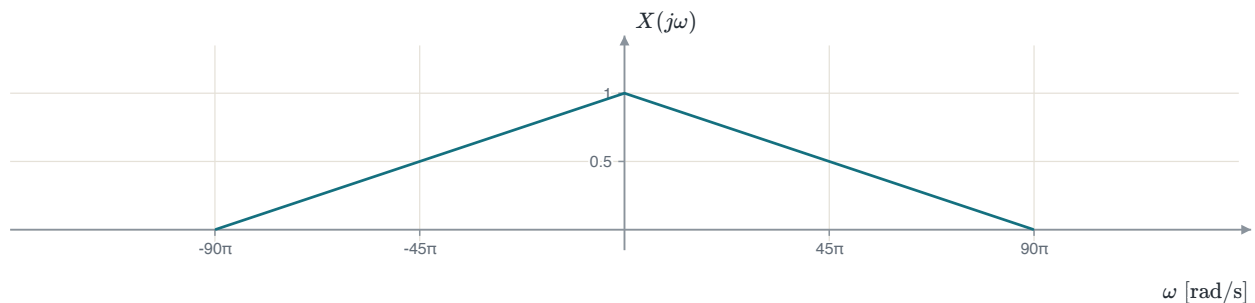
A signal has the triangular spectrum shown below, with $X(j\omega) = 0$ for $|\omega| > 50\pi$ rad/s. It is sampled with period $T = 1/300$ s.



- Read ω_M off the figure.
- Give ω_s and the guard band $\omega_s - 2\omega_M$.
- Sketch $X_p(j\omega)$ for $k = -1, 0, 1$, labelling both positive and negative frequencies.

D7-14

A signal has the triangular spectrum shown below, with $X(j\omega) = 0$ for $|\omega| > 90\pi$ rad/s. It is sampled at $\omega_s = 140\pi$ rad/s.



- Confirm that this is undersampling.
- Sketch the baseband together with the neighbouring copies, shading the overlap.
- Give the width of the overlap on each side.

D7-15

The signal $x(t) = \cos(2\pi \cdot 450t)$ is sampled at $f_s = 800$ Hz.

- State whether the sampling theorem is satisfied.
- Give the apparent frequency of the sampled sequence.
- Sketch the spectral lines involved: the original frequency and the apparent one.

D7-16

The signal

$$x(t) = \cos(2\pi \cdot 150t) + \cos(2\pi \cdot 350t)$$

is sampled at $f_s = 500$ Hz.

- Give the apparent frequency of each component.
- Show that the two components alias onto the same apparent frequency, and simplify the sampled sequence.
- Besides 350 Hz, name one further frequency above 500 Hz whose samples at this rate coincide with those of the 150 Hz component.

D7-17

A signal is sampled with period $T = 0.4$ ms and reconstructed with an ideal filter of cutoff $\omega_c = \pi/T$. The only non-zero samples are $x(0) = 3$ and $x(T) = -2$.

- Write $h_{LP}(t)$ in closed form, using the unnormalised sinc convention $\text{sinc}(\theta) = \sin \theta / \theta$.
- Write $x_r(t)$ as a sum of two shifted kernels.
- Evaluate $x_r(T)$ directly from the formula, and confirm it matches the given sample.
- Evaluate $x_r(1.5T)$.

D7-18

A zero-order hold has sampling period $T = 0.25$ ms.

- Give $H_0(j\omega)$ in closed form.
- Evaluate $H_0(0)$ and compare it with the ideal reconstruction filter's gain.
- Evaluate $|H_0(j\pi/T)|$, and give it as a fraction of the ideal gain.

D7-19

The signal $x(t) = \cos(50\pi t) + \cos(140\pi t)$ is sampled at $f_s = 100$ Hz and reconstructed with cutoff $\omega_c = \omega_s/2$.



- Without any filter ahead of the sampler, give the reconstructed signal and the error $e(t) = x(t) - x_r(t)$.
- With a lowpass filter of cutoff $\omega_s/2$ placed ahead of the sampler, give the reconstructed signal and the error.
- Compute the time-averaged mean-square error in both cases and compare them.

D7-20

Let

$$x(t) = 5 + 3 \cos(1200\pi t) + 6 \sin(3600\pi t),$$

sampled exactly at its own Nyquist rate.

- a) Give ω_M and confirm $\omega_s = 2\omega_M = 7200\pi$ rad/s, so $T = 1/3600$ s.
- b) Show directly from the samples that the $\sin(3600\pi t)$ term contributes nothing.
- c) Show the same result in the frequency domain, from the weights the baseband and the $k = 1$ copy carry at $\omega = 3600\pi$.
- d) A guard band of $\omega_g = 400\pi$ rad/s is now required. Give the new minimum sampling rate.

D7-21

Determine the Nyquist sampling rate for each of the following signals.

- a) $y(t) = x(t) + x^2(t)$, where $x(t) = \cos(80\pi t) + \cos(160\pi t)$.
- b) $y(t) = \frac{\sin(80\pi t)}{\pi t} [1 + e^{j200\pi t}]$.
- c) $y(t) = \frac{\sin(80\pi t)}{\pi t} * \frac{\sin(240\pi t)}{\pi t}$, where $*$ is the convolution operator.

D7-22

Plot the Fourier transforms of each of the signals given below and determine the Nyquist sampling frequencies (in rad/s) for each of these signals.

- a) $x(t) = \frac{\sin(30\pi t)}{\pi t} + \cos(40\pi t)$.
- b) $x(t) = \cos(30\pi t) \cos(70\pi t)$.
- c) $x(t) = \frac{\sin(30\pi t)}{\pi t} \frac{\sin(60\pi t)}{\pi t}$.

D7-23

Plot the Fourier transforms of each of the signals given below, and determine the Nyquist sampling frequencies (in rad/s) for each of these signals while considering a guard band of $\omega_g = 80\pi$ rad/s.

- a) $x(t) = \frac{\sin(120\pi t)}{\pi t} + \frac{\sin(240\pi t)}{\pi t}$.
- b) $h(t) = x(t) \cos(160\pi t)$.
- c) $y(t) = x(t) * h(t)$, where $*$ is the convolution operator.

D7-24

A signal $x(t) = 3 \cos(200\pi t) + 2 \sin(500\pi t)$ is sampled at $f_s = 400$ Hz.

- a) Determine the Nyquist rate of $x(t)$ in hertz.
- b) State which components alias at $f_s = 400$ Hz, and give the apparent frequency of each.
- c) Write the continuous-time signal that the samples appear to have come from.

D7-25

A band-limited signal has $X(j\omega) = 0$ for $|\omega| > 1000\pi$ and is sampled with an impulse train of period $T = \frac{1}{1500}$ s.

- a) Give the sampling frequency ω_s and say whether the Nyquist condition is met.
- b) Describe the spectrum of the sampled signal $x_p(t)$ and the width of the gap between copies.
- c) Give an ideal reconstruction filter that recovers $x(t)$ exactly, with its gain and a valid cut-off.

D7-26

An audio channel is sampled at $f_s = 8$ kHz with no anti-aliasing filter. Four pure tones are applied in turn: 3 kHz, 5 kHz, 9 kHz and 12 kHz.

- Give the folding frequency and say which tones are sampled correctly.
- Give the apparent frequency of each tone after sampling.
- Two of the four tones become indistinguishable. Identify them and say what filter placed where would have prevented it.

D7-27

Determine the Nyquist sampling rate for each of the following signals.

- $y(t) = \cos^2(300\pi t)$.
- $y(t) = \frac{\sin(100\pi t)}{\pi t} \cos(400\pi t)$.
- $y(t) = \left[\frac{\sin(150\pi t)}{\pi t} \right]^2$.

D7-28

A signal $x(t)$ is band-limited to $|\omega| < 600\pi$ and is sampled at $\omega_s = 2000\pi$ rad/s to give the sequence $x[n] = x(nT)$. Every second sample is then discarded, giving $v[n] = x[2n]$.

- Give the effective sampling rate after the samples are discarded.
- State whether $x(t)$ can still be recovered from $v[n]$, with reasons.
- Give the largest bandwidth a signal could have and still survive the same treatment.

D7-29

Determine the Nyquist sampling rate for each of the following signals, where $\omega_g = 0$ (no guard band is required).

- a) $y(t) = x_1(t)x_2(t)$, where x_1 is band-limited to 200π and x_2 to 500π .
- b) $y(t) = x_1(t) + x_2(t)$, with the same two signals.
- c) $y(t) = x_1(t) * x_2(t)$, with the same two signals.

D7-30

A signal $x(t) = \frac{\sin(400\pi t)}{\pi t}$ is sampled at $\omega_s = 1000\pi$ rad/s, and the samples are passed through an ideal reconstruction filter of gain T and cut-off $\omega_c = 500\pi$.

- a) Give $X(j\omega)$ and the Nyquist rate of $x(t)$.
- b) Describe the spectrum after sampling and say whether the copies overlap.
- c) Give the output of the reconstruction filter, and repeat the three parts for $\omega_s = 600\pi$.



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v0	23 September 2026	First edition.

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